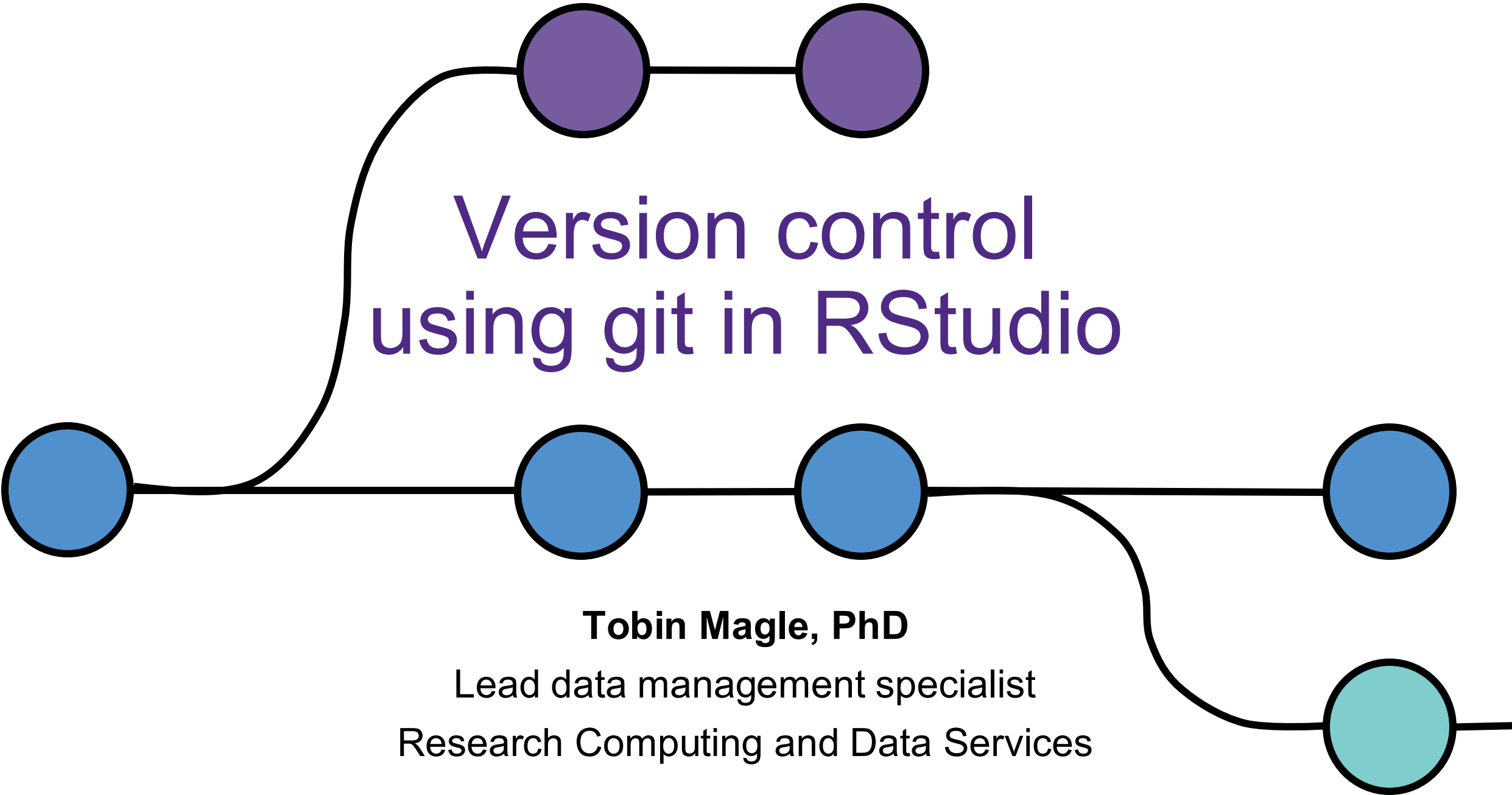




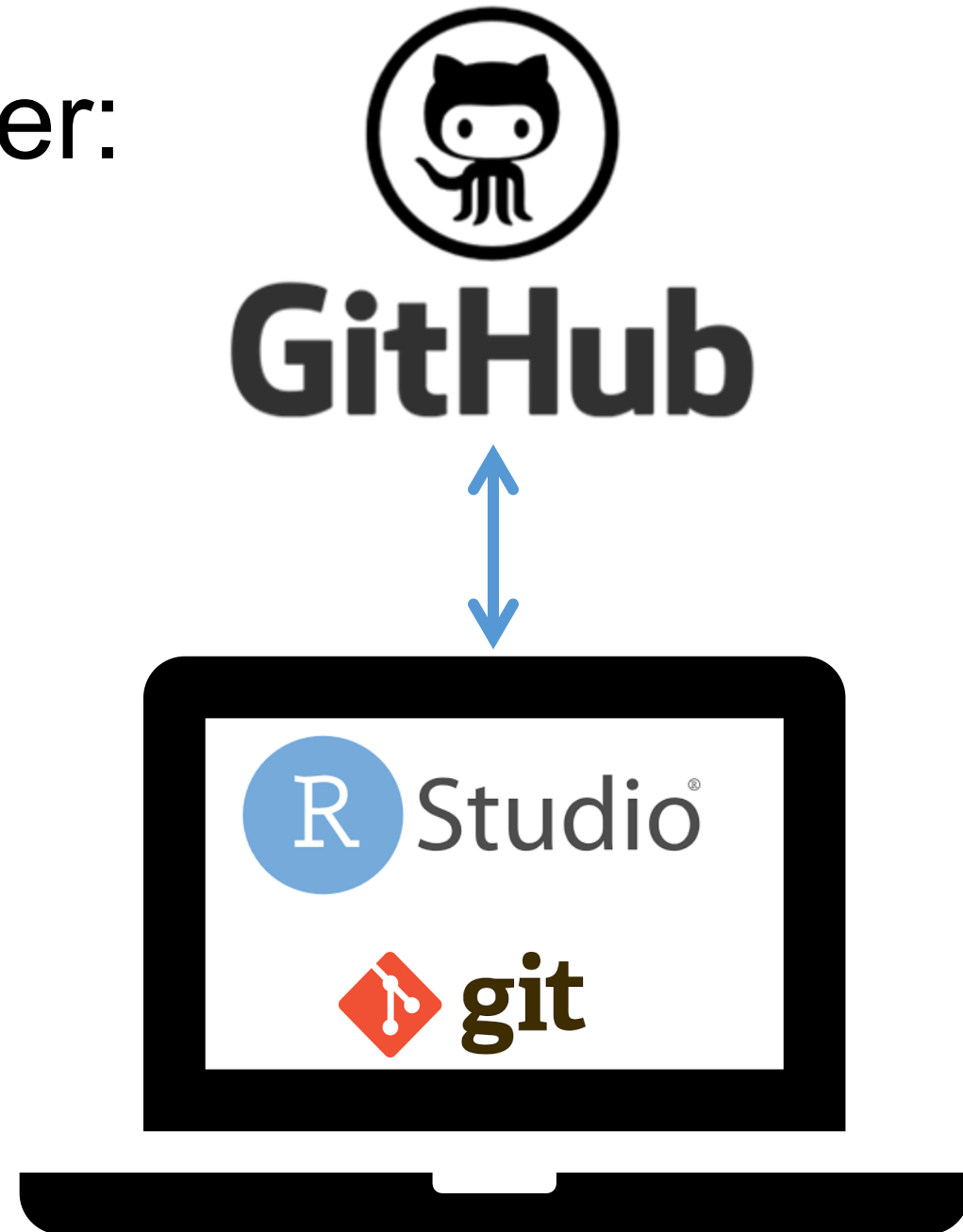
Version control using git in RStudio

Tobin Magle, PhD

Lead data management specialist
Research Computing and Data Services

Today we're going to cover:

- What is **version control**?
- What is **git**?
- Version control using **RStudio**
- Backing up your data to **GitHub**



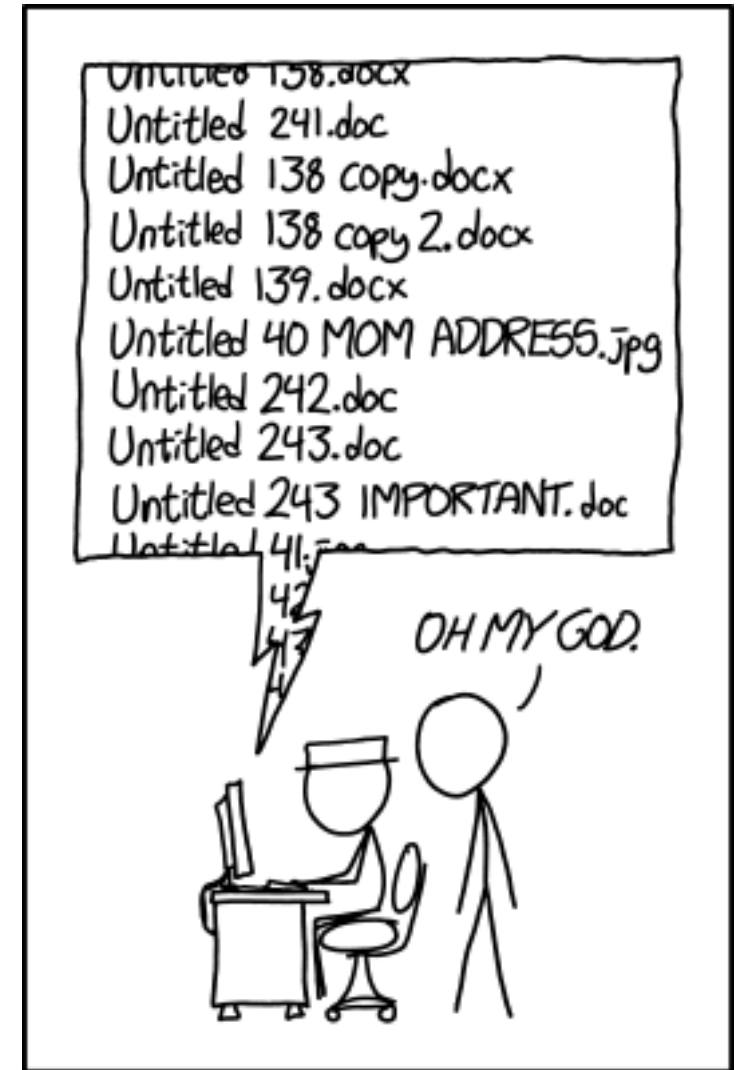
Pre-workshop Setup

- Install R installed
- Install RStudio
- Install git
- Created a GitHub account

Manual Version Control

“Renaming files”

- Creates many files
- Relies on your ability to name files consistently
- Not designed for collaboration



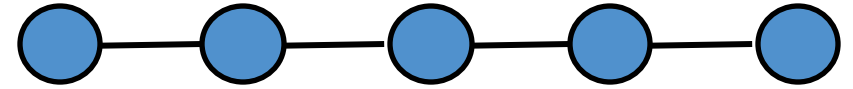
PROTIP: NEVER LOOK IN SOMEONE ELSE'S DOCUMENTS FOLDER.

Version Control Systems

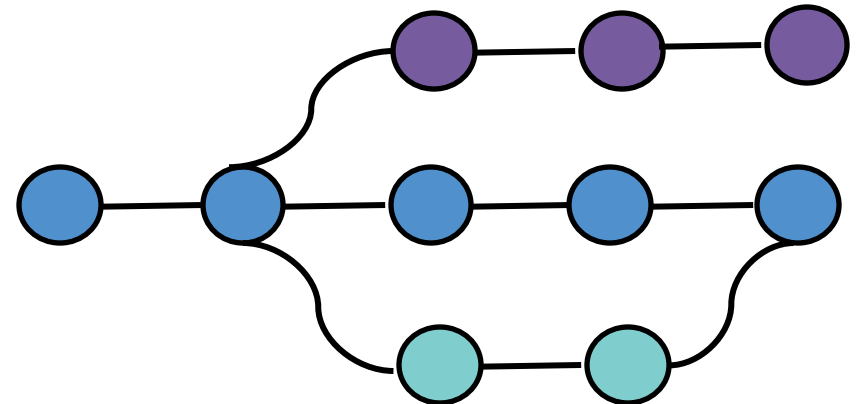
“Unlimited undo”

- One file, many versions
- Records who made what change when
- Good for **collaboration**: identifies and helps resolve conflicting changes

Solo

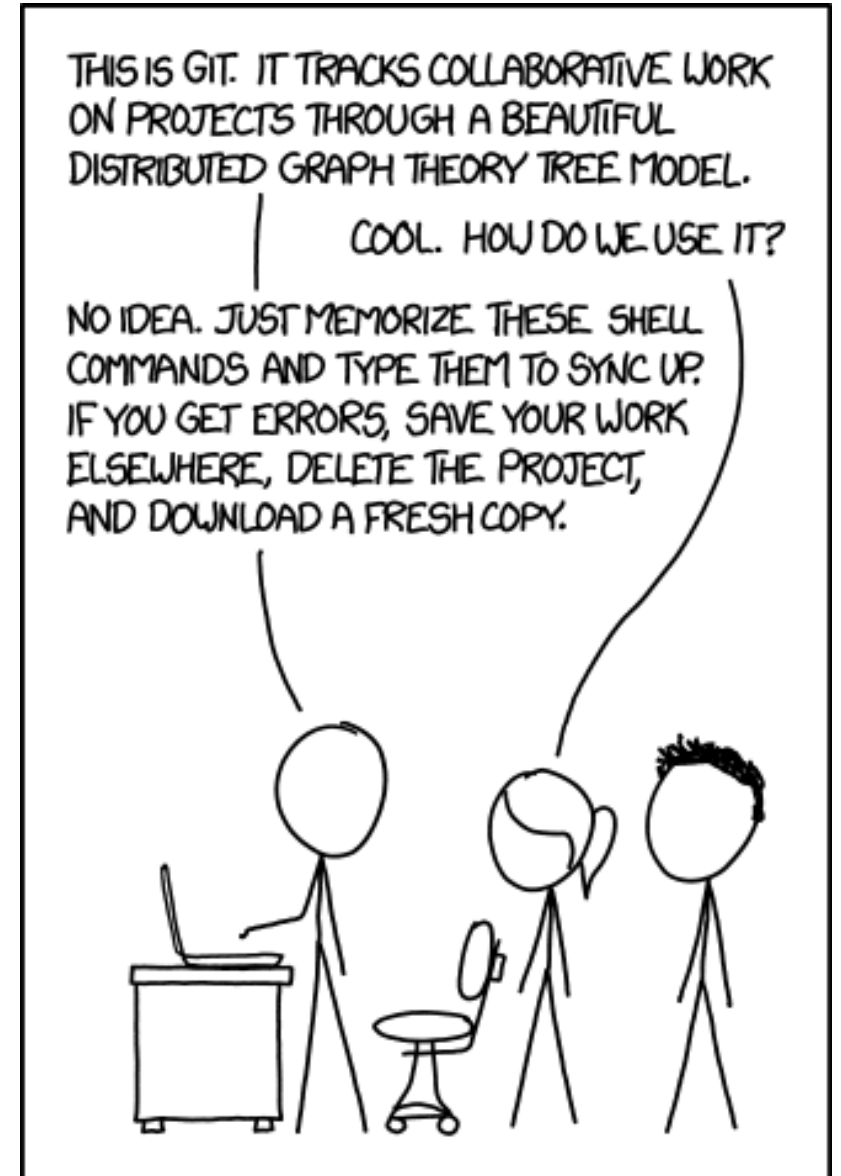


Collaborative



Today we're using git

- Built for large software development projects
- Can get complicated, but the basics are simple
- Free, open source
- Many online resources/tutorials



Types of files git works best with

“Good” formats (text based)

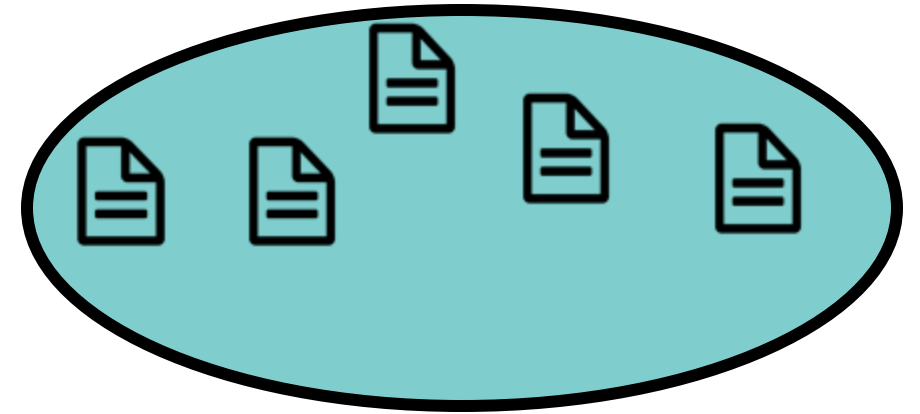
- Documents (.txt, .tex, .rtf, .Rmd)
- Tabular Data (.csv)
- Source code (.R, .py, .c, .sh)

“Bad” formats (binary files)

- Documents (.docx, .pdf, .ppt)
- Excel spreadsheets (.xlsx)
- Media (.jpg, .mp3, .mp4)
- Databases (.mdb, .sqlite)

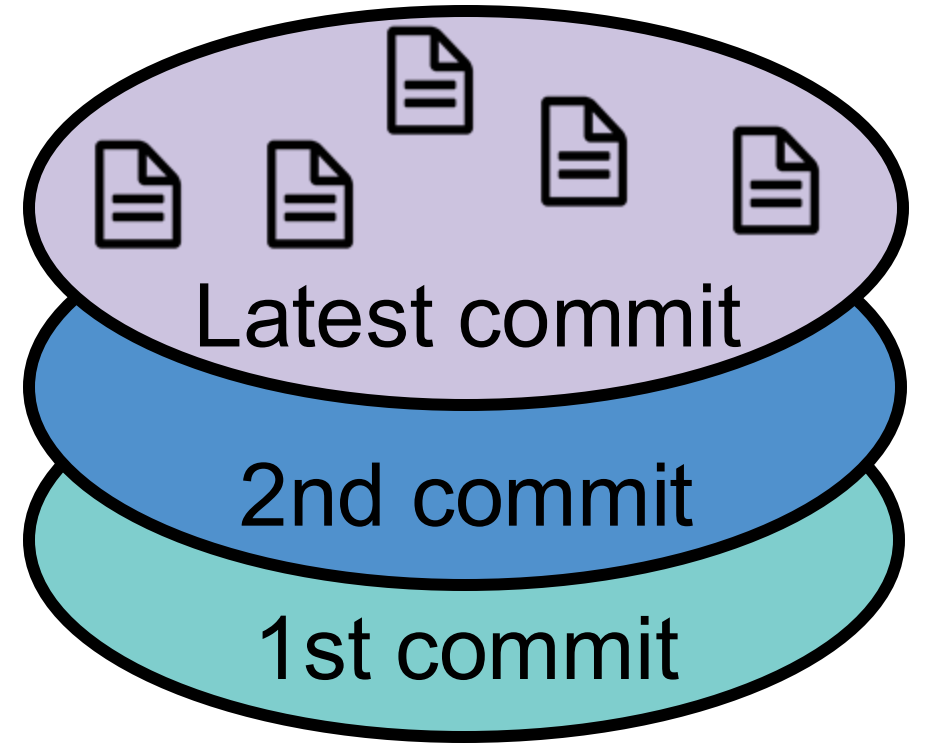
Using git creates a repository

- **Repository** – A folder that is tracked by git



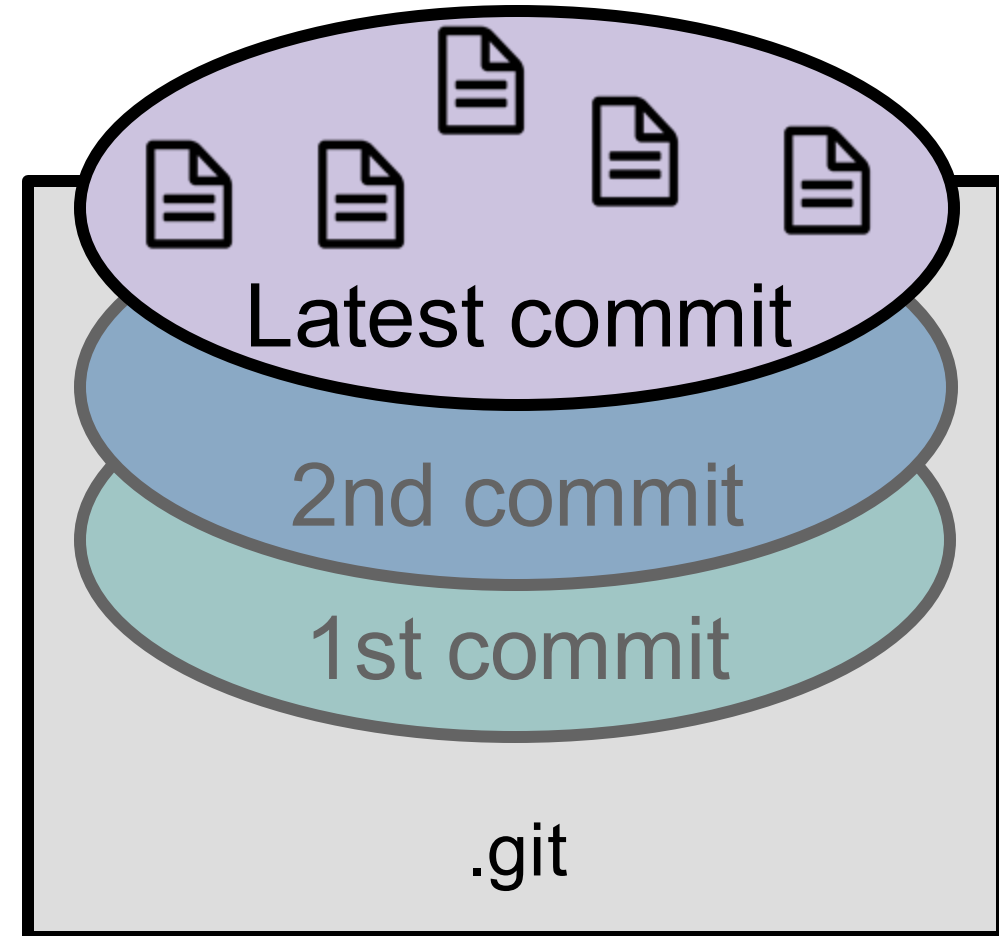
Repositories are made of commits

- **Repository** – A folder that is tracked by git
- **Commit** – a version of what the folder looks like when it was created



You only see your latest commit

- **Repository** – A folder that is tracked by git
- **Commit** – a version of what the folder looks like when it was created
- **Structure is hidden** (.git), you see the most recent version



Many ways to use Git

Command line

```
C:\Windows\System32\cmd.exe

D:\GFG>git status
On branch master

Initial commit

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)

    new file:   Basic Code/C++ Project.cbp
    new file:   Basic Code/README.md
    new file:   Basic Code/bin/Debug/C++ Project.exe
    new file:   Basic Code/data.in
    new file:   Basic Code/data.out
    new file:   Basic Code/main.cpp
    new file:   Basic Code/obj/Debug/main.o
    new file:   DS/stackPar.cpp
    new file:   DS/stackPar.exe
    new file:   DS/stackPar.o
    new file:   DS/stackPar.pdb
```

Desktop Clients

Current branch: **esc-pr** #3972 ✓

Fetch origin
Last fetched 2 minutes ago

Add event handler to dropdown component

iAmWillShepherd and Markus Olsson committed c79e71c 1 changed file

Co-Authored-By: Markus Olsson <niik@users.noreply.github.com>

File	Line	Code
app\src\ui\toolbar\dropdown.tsx	145	@@ -145,6 +145,10 @@ export class ToolbarDropdown extends React.Component<
	146	this.state = { clientRect: null }
	147	}
	148	+ private get isOpen() {
	149	+ return this.props.dropdownState === 'open'

GitHub Website

Edited the readme file

Changed the title to something better. Added a bulleted list

main

maglet committed 38 minutes ago Verified 1 parent 20225f0 commit e1b19cd

Showing 1 changed file with 5 additions and 2 deletions.

Split Unified

7 README.md

Line	Old	New
1	- # github-gui-demo	+ # Demo repository for using the GitHub GUI
2		

RStudio IDE

Environment History Connections Git Tutorial

Diff Commit Pull Push History More

main

Staged	Status	Path
☐	M	script.R
☐	?	data/2023-08-25-Globus_Usage_Transfer_Detail.csv
☐	M	data/Globus_Usage_Transfer_Detail.csv
☐	M	data/RDSS_users.csv

Many ways to use Git

Command line

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C:\Windows\System32\cmd.exe

D:\ESC>git status
On branch master

Initial commit

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)

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        new file:   Basic Code/README.md
        new file:   Basic Code/bin/Debug/C++ Project.exe
        new file:   Basic Code/data.in
        new file:   Basic Code/data.out
        new file:   Basic Code/main.cpp
        new file:   Basic Code/obj/Debug/main.o
        new file:   DS/stackPar.cpp
        new file:   DS/stackPar.exe
        new file:   DS/stackPar.o
        new file:   DS/stackPar.cpp
```

Desktop Clients

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app\src\ui\toolbar\dropdown.tsx			
	145	145	@@ -145,6 +145,10 @@ export class ToolbarDropdown extends React.Component<
	146	146	this.state = { clientRect: null }
	147	147	}
		148	+ private get isOpen() {
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Split Unified

7 README.md

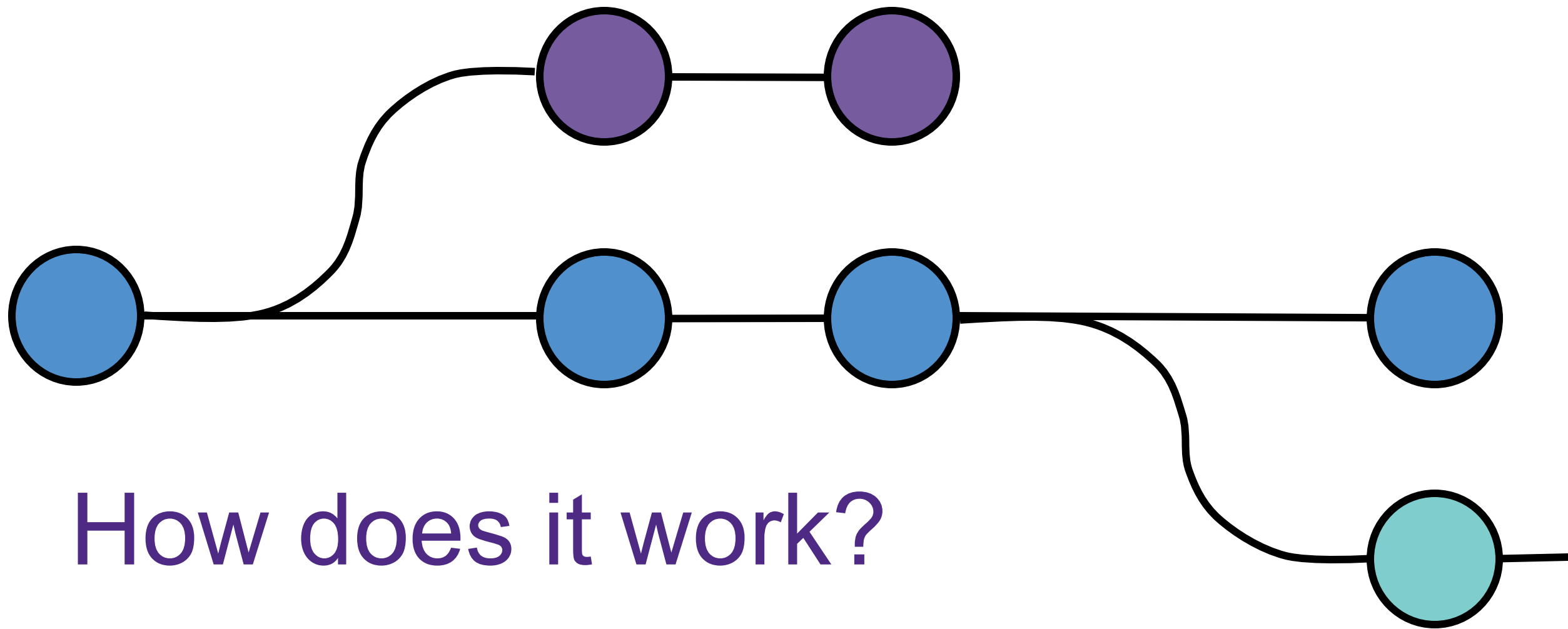
...	@@ -1,3 +1,6 @@
1 - # github-gui-demo	1 + # Demo repository for using the GitHub GUI
2	2

RStudio IDE

Environment History Connections Git Tutorial

Diff Commit Pull Push History More main

Staged	Status	Path
☐	M	script.R
☐	? ?	data/2023-08-25-Globus_Usage_Transfer_Detail.csv
☐	M	data/Globus_Usage_Transfer_Detail.csv
☐	M	data/RDSS_users.csv



How does it work?

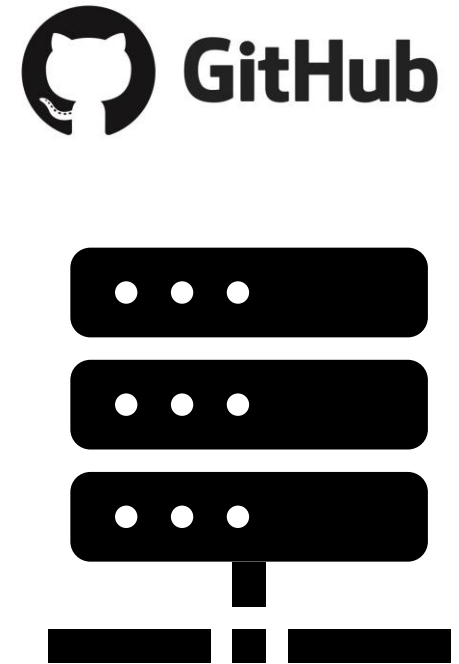
Coding happens on your computer



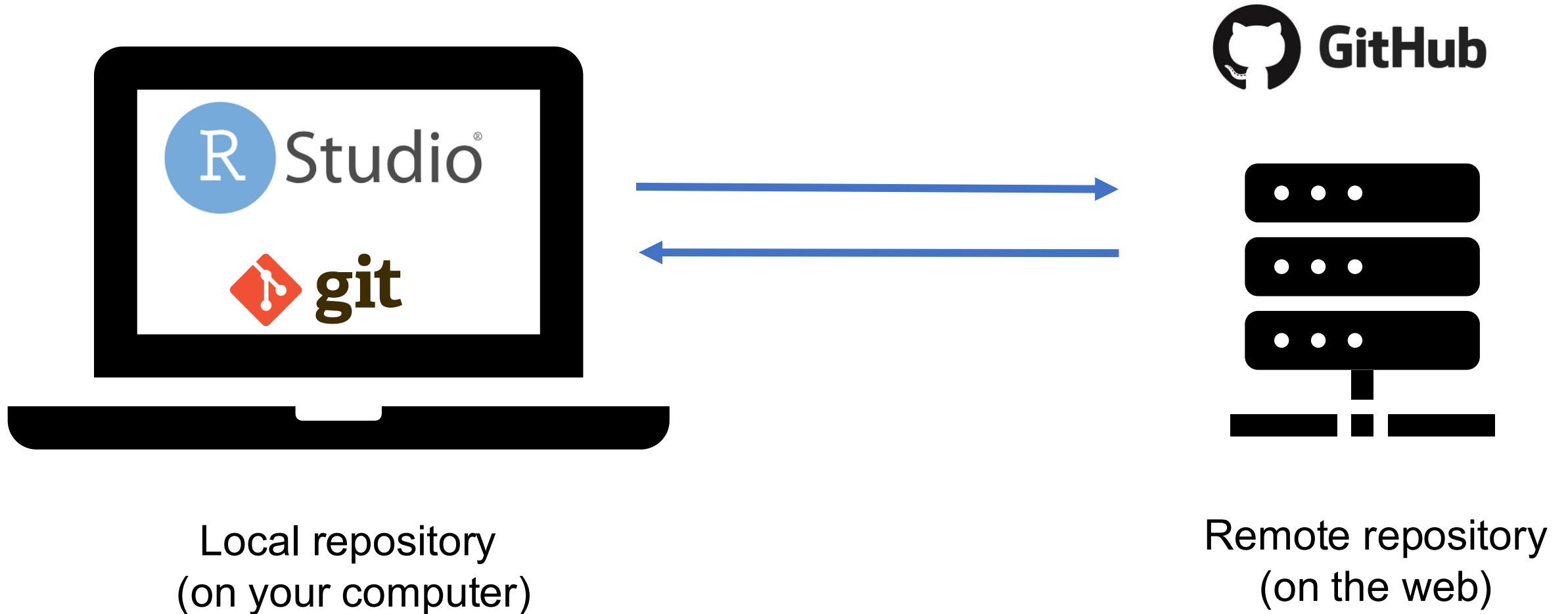
- Create and edit files in RStudio
- Git is also on your computer
- Use git version control in the RStudio interface
- Creates a local repository

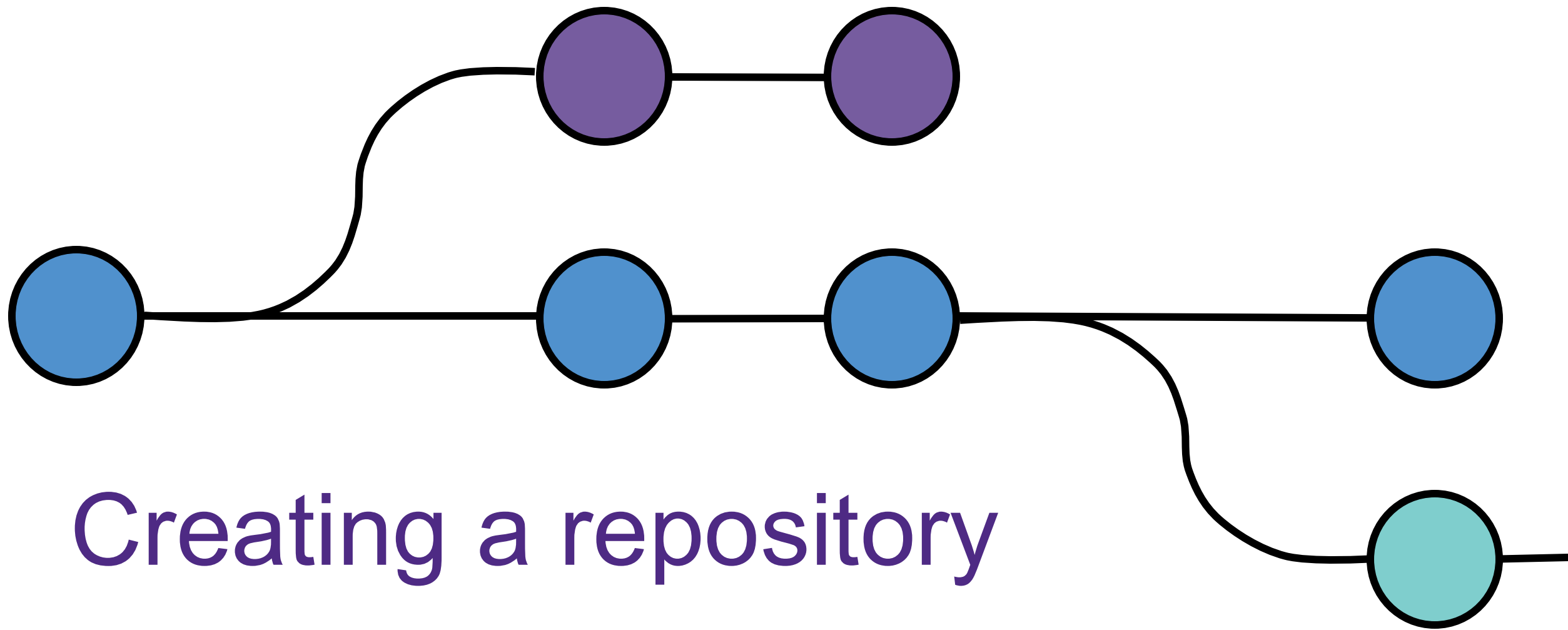
GitHub stores code in the cloud

- Back up your work
- Collaborate with colleagues
- Share your code



Overall workflow

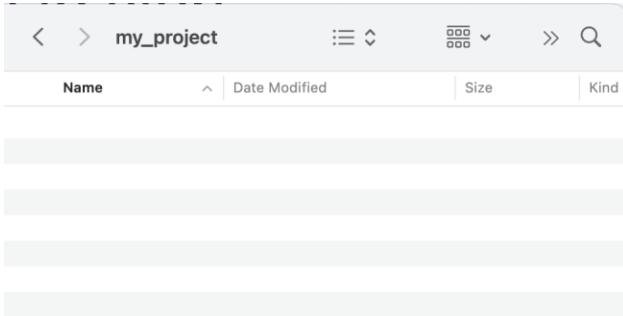




Creating a repository

A few ways to start

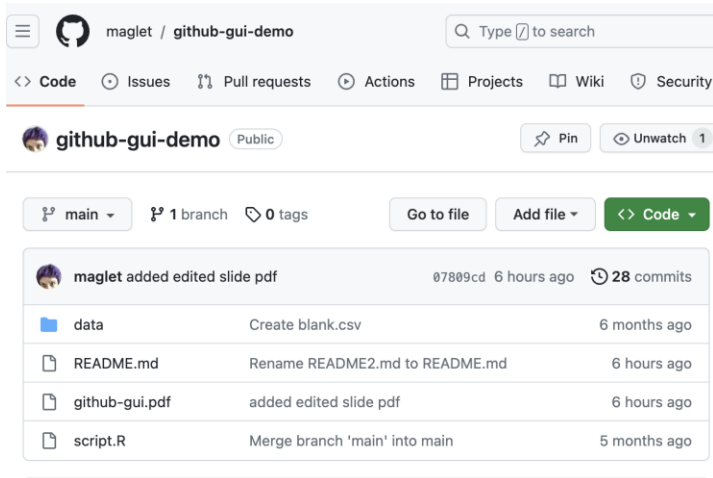
From nothing



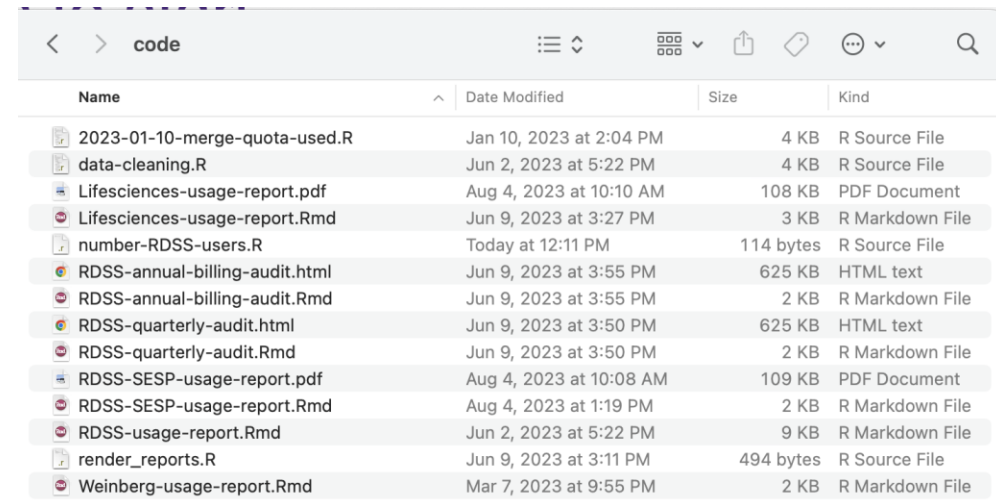
From a local repo

```
(base) EDMRCSLFYLNWQ0KJ:globus-stats tobinmagle$ ls -la
.
..
.DS_Store
.RData
.Rhistory
.Rproj.user
.git
.gitignore
globus-stats.Rproj
script.R
(base) EDMRCSLFYLNWQ0KJ:globus-stats tobinmagle$
```

From a GitHub repo

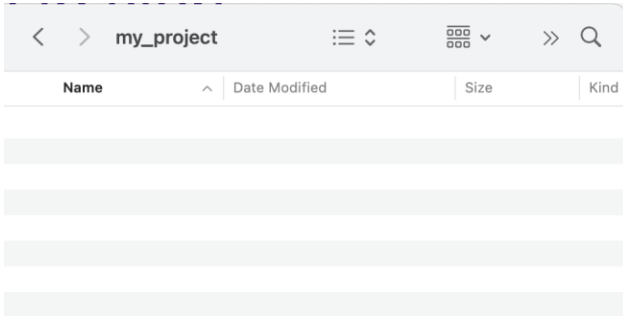


From files outside a repo



A few ways to start

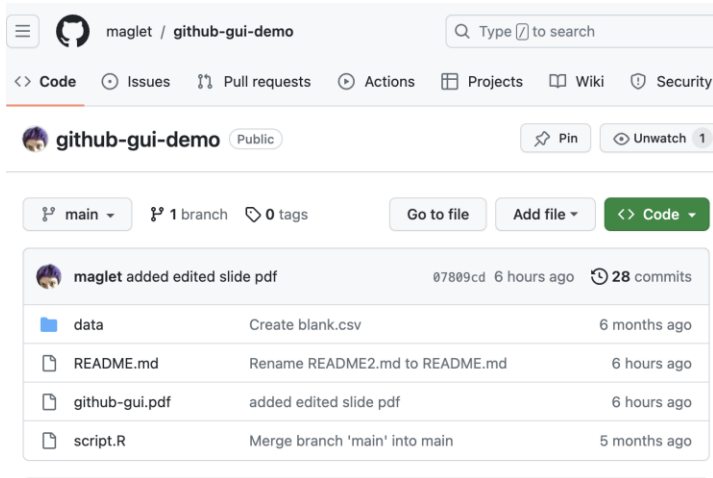
From nothing



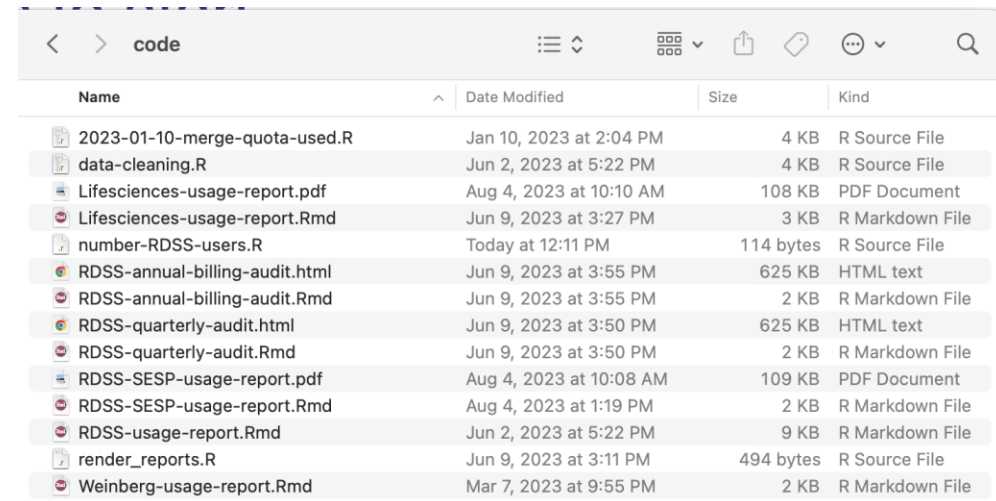
From a local repo

```
(base) EDMRCSLFYLNWQ0KJ:globus-stats tobinmagle$ ls -la
.
..
.DS_Store
.RData
.Rhistory
.Rproj.user
.git
.gitignore
globus-stats.Rproj
script.R
(base) EDMRCSLFYLNWQ0KJ:globus-stats tobinmagle$
```

From a GitHub repo

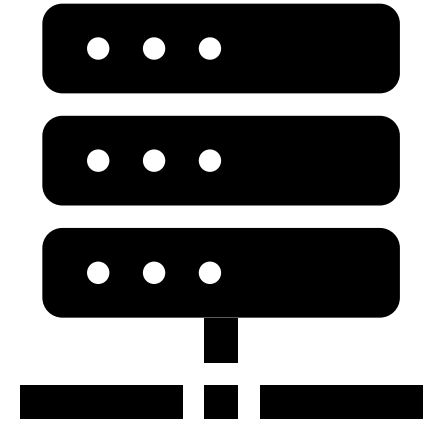


From files outside a repo



Name	Date Modified	Size	Kind
2023-01-10-merge-quota-used.R	Jan 10, 2023 at 2:04 PM	4 KB	R Source File
data-cleaning.R	Jun 2, 2023 at 5:22 PM	4 KB	R Source File
Lifesciences-usage-report.pdf	Aug 4, 2023 at 10:10 AM	108 KB	PDF Document
Lifesciences-usage-report.Rmd	Jun 9, 2023 at 3:27 PM	3 KB	R Markdown File
number-RDSS-users.R	Today at 12:11 PM	114 bytes	R Source File
RDSS-annual-billing-audit.html	Jun 9, 2023 at 3:55 PM	625 KB	HTML text
RDSS-annual-billing-audit.Rmd	Jun 9, 2023 at 3:55 PM	2 KB	R Markdown File
RDSS-quarterly-audit.html	Jun 9, 2023 at 3:50 PM	625 KB	HTML text
RDSS-quarterly-audit.Rmd	Jun 9, 2023 at 3:50 PM	2 KB	R Markdown File
RDSS-SESP-usage-report.pdf	Aug 4, 2023 at 10:08 AM	109 KB	PDF Document
RDSS-SESP-usage-report.Rmd	Aug 4, 2023 at 1:19 PM	2 KB	R Markdown File
RDSS-usage-report.Rmd	Jun 2, 2023 at 5:22 PM	9 KB	R Markdown File
render_reports.R	Jun 9, 2023 at 3:11 PM	494 bytes	R Source File
Weinberg-usage-report.Rmd	Mar 7, 2023 at 9:55 PM	2 KB	R Markdown File

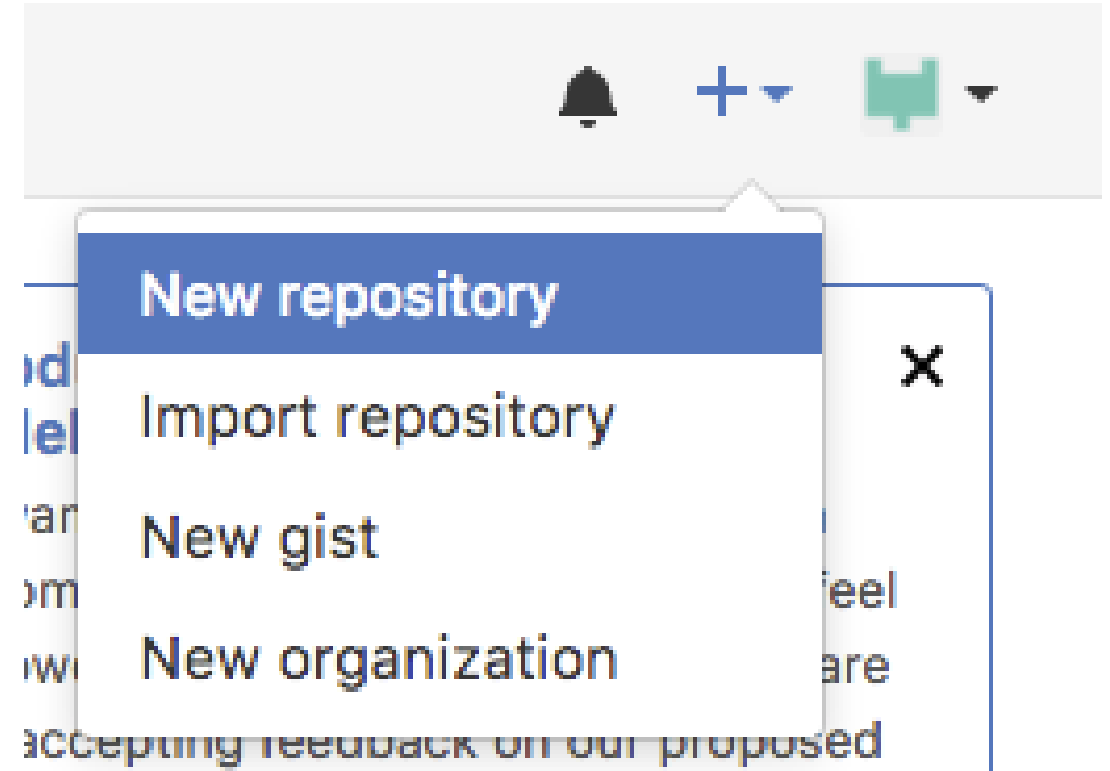
Start in GitHub



Remote

GitHub: Create a repository

1. Go to <http://github.com>
2. Click the plus sign on the upper right-hand side of the screen
3. Select **New repository**



GitHub: Create a repository

A new page will open:

- Choose an owner
- Name your repository
- Provide a description
- Keep it public
- Add a README file

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner *  maglet ▾ / **Repository name ***

✔ name-your-repo is available.

Great repository names are short and memorable. How about [studious-goggles?](#)

Description

0 / 350 characters

2 Configuration

Choose visibility *
Choose who can see and commit to this repository Public ▾

Start with a template
Templates pre-configure your repository with files. No template ▾

Add README
READMEs can be used as longer descriptions. [About READMEs](#) Off ☐

Add .gitignore
.gitignore tells git which files not to track. [About ignoring files](#) No .gitignore ▾

Add license
Licenses explain how others can use your code. [About licenses](#) No license ▾

[Create repository](#)

GitHub: Choose an Owner

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).

Required fields are marked with an asterisk ().*

1 General

Owner *
 maglet ▾

Repository name *

✓ name-your-repo is available.

- GitHub repositories can be associated with users or "orgs"
- For this workshop, select your Git username
- If your research group has an org, use that for your research code

GitHub: Name your repository

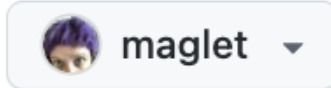
Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).

Required fields are marked with an asterisk ().*

1 General

Owner *



Repository name *

name-your-repo

✓ name-your-repo is available.

- Make a descriptive name for your repo
- For this workshop, use your last name then "git-workshop"
- Eg: "magle-git-workshop"

GitHub: Provide a description

Great repository names are short and memorable. How about **studious-goggles**?

Description

0 / 350 characters

- Write a short description for what this repo is for
- Example: "Test repo for a workshop on using Git in RStudio"
- The descriptions for your research code will be more informative

GitHub: Keep it public

2

Configuration

Choose visibility *

Choose who can see and commit to this repository

 Public ▼

- Keep the repo public for this workshop
- That way we don't have to worry about configuration till we upload changes
- For your research code, you might want to keep it private

GitHub: Add a README file

Add README
READMEs can be used as longer descriptions. [About READMEs](#)

Add .gitignore
.gitignore tells git which files not to track. [About ignoring files](#)

Add license
Licenses explain how others can use your code. [About licenses](#)

Off ☐

No .gitignore ▼

No license ▼

- A **README file** contains the repo description and any other information that is useful to know about your code.
- Preview of this file will be on the main repo page
- We will edit this file during the workshop to show how to save changes to text files using git.

GitHub: Create repository

Add README
READMEs can be used as longer descriptions. [About READMEs](#)

Add .gitignore
.gitignore tells git which files not to track. [About ignoring files](#)

Add license
Licenses explain how others can use your code. [About licenses](#)

Off ☐

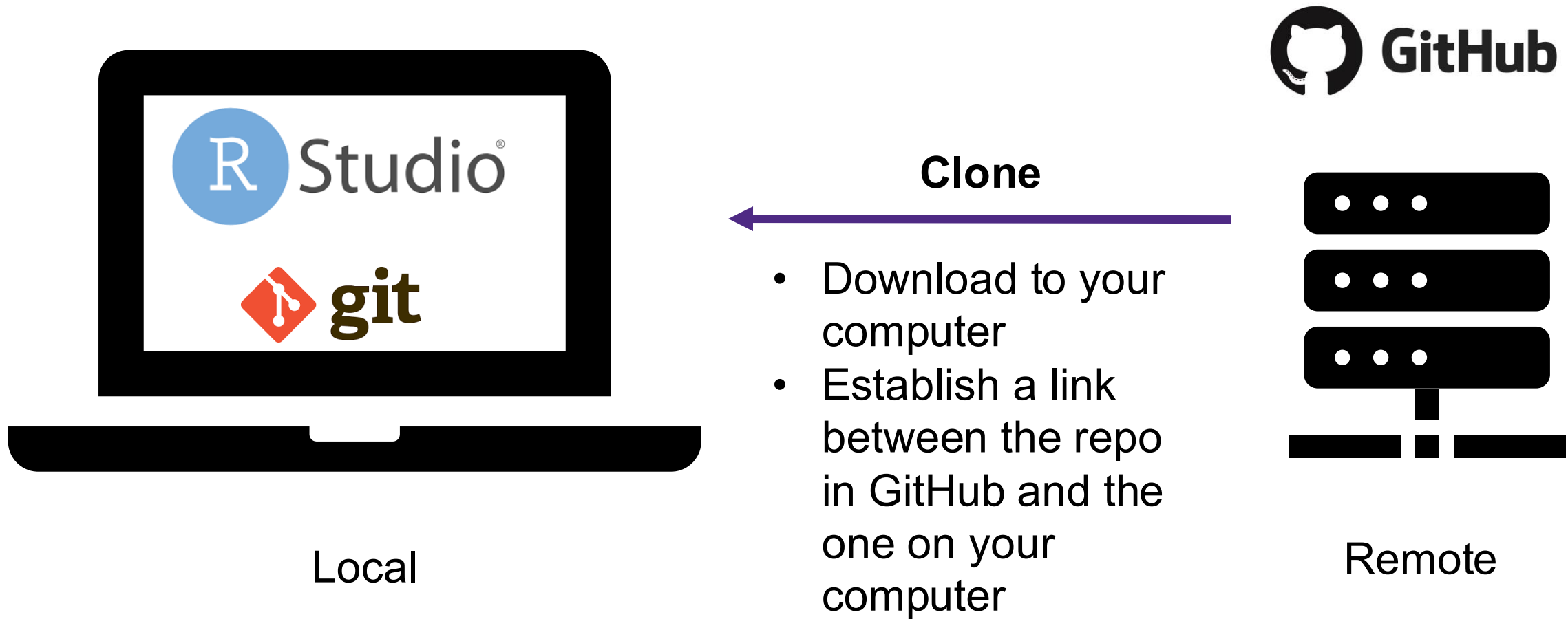
No .gitignore ▼

No license ▼

Create repository

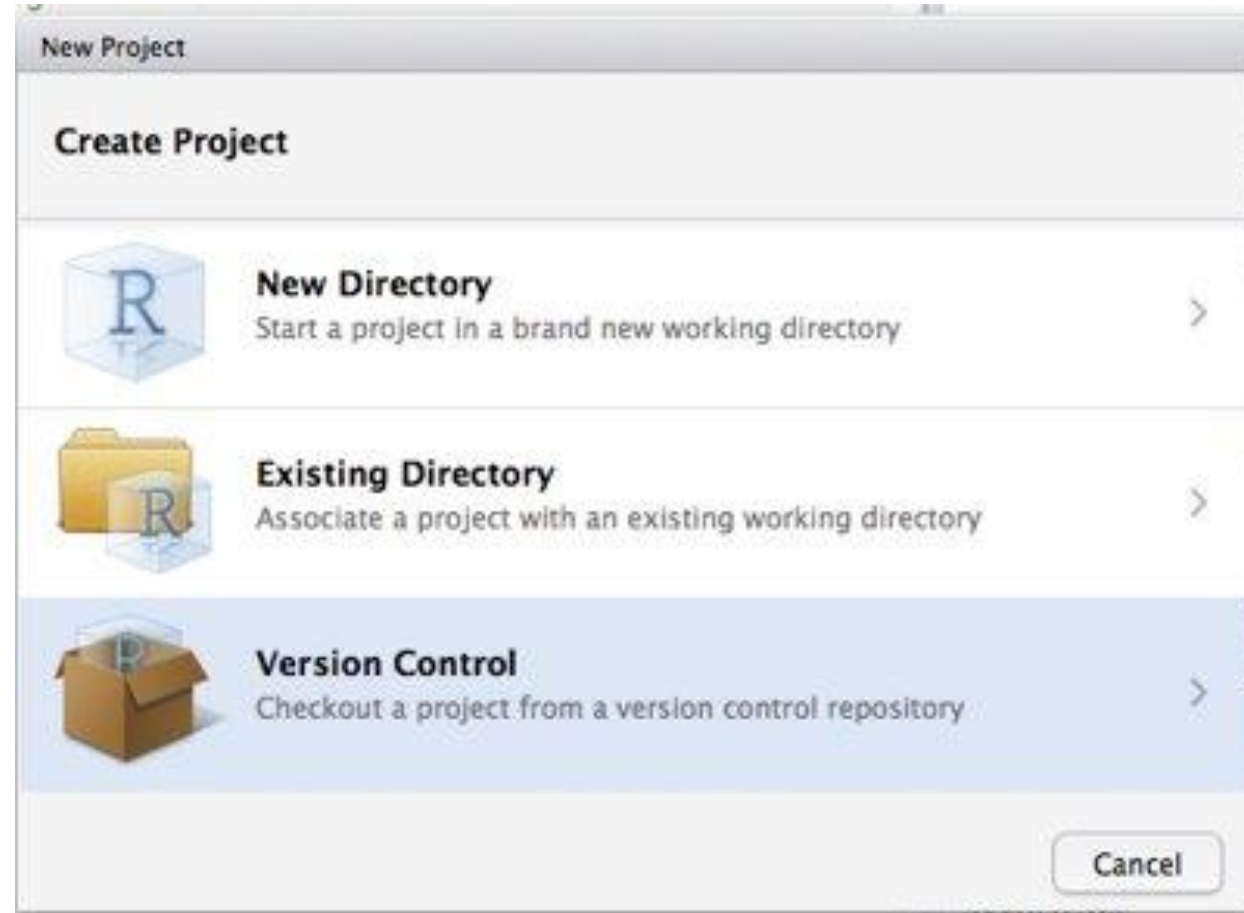
- Creates a repository (<https://github.com/REPONAME>)
- Visible to anyone
- Contains a README with the description you entered previously

Clone to your computer



RStudio: Make a New Project

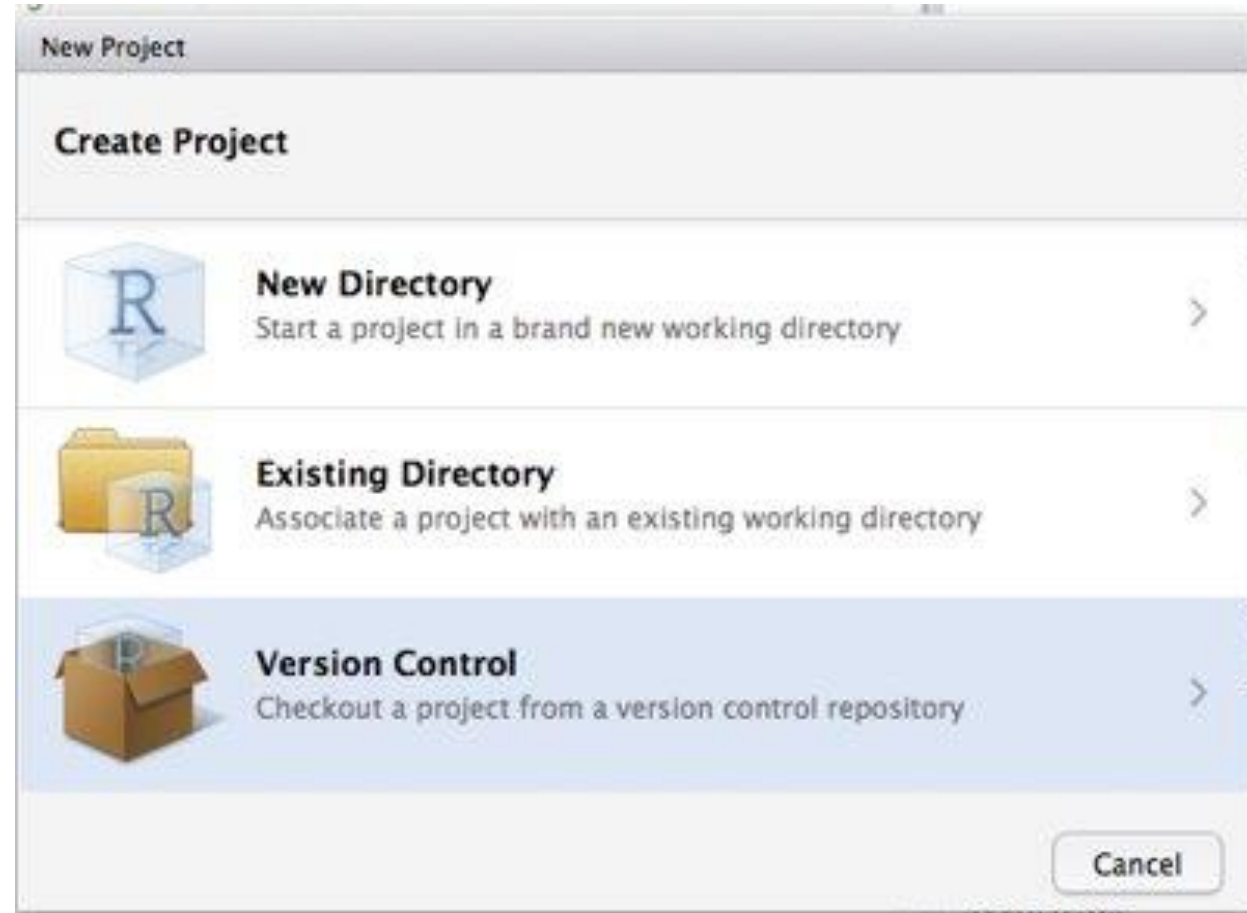
- RStudio projects are very useful for reproducibility
- They also make setting up a GitHub repository from Git very easy



RStudio: Make a New Project

To create a new RStudio Project

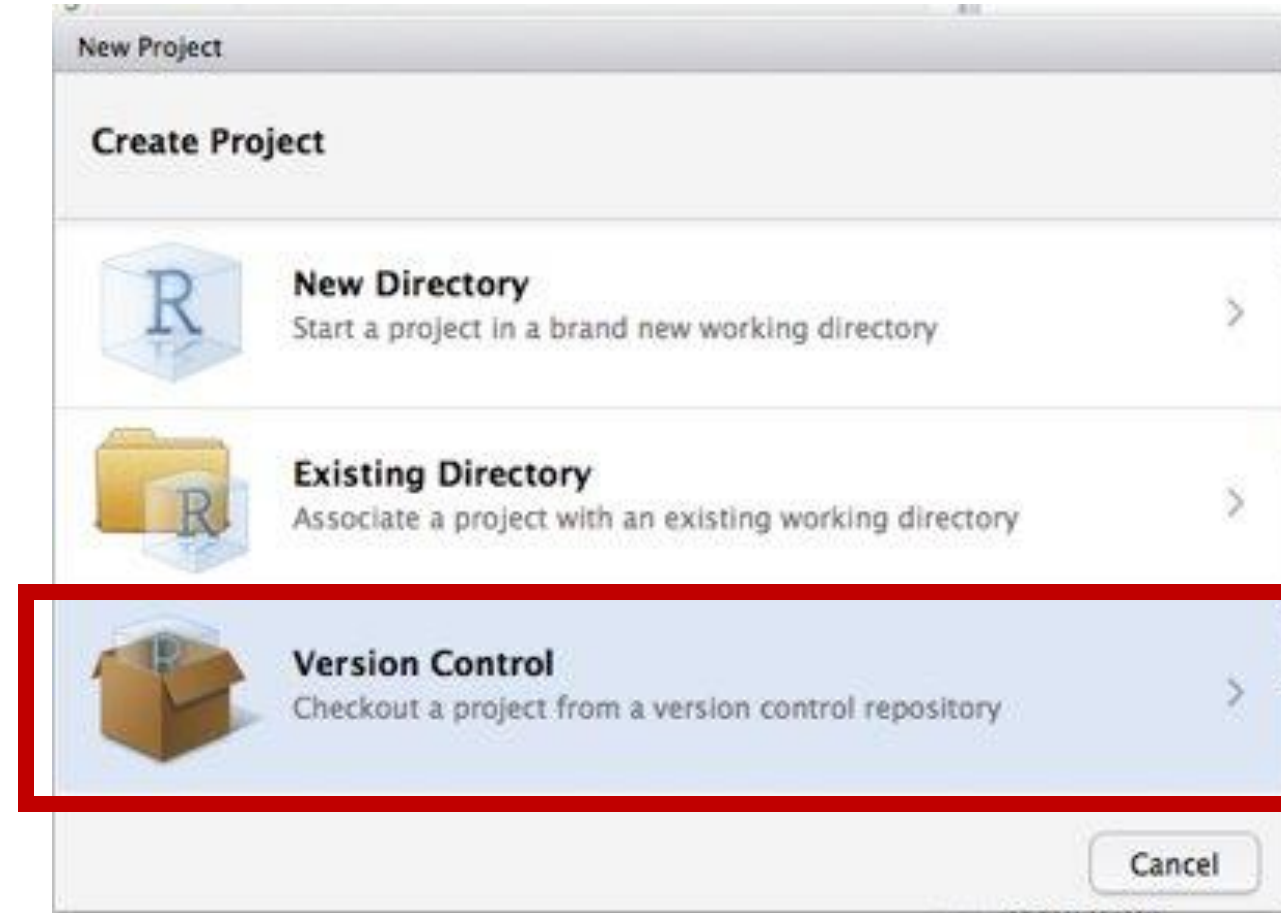
- Select the **File** menu
- Select **New Project**
- A new window will open



RStudio: Make a New Project

To create a new RStudio Project

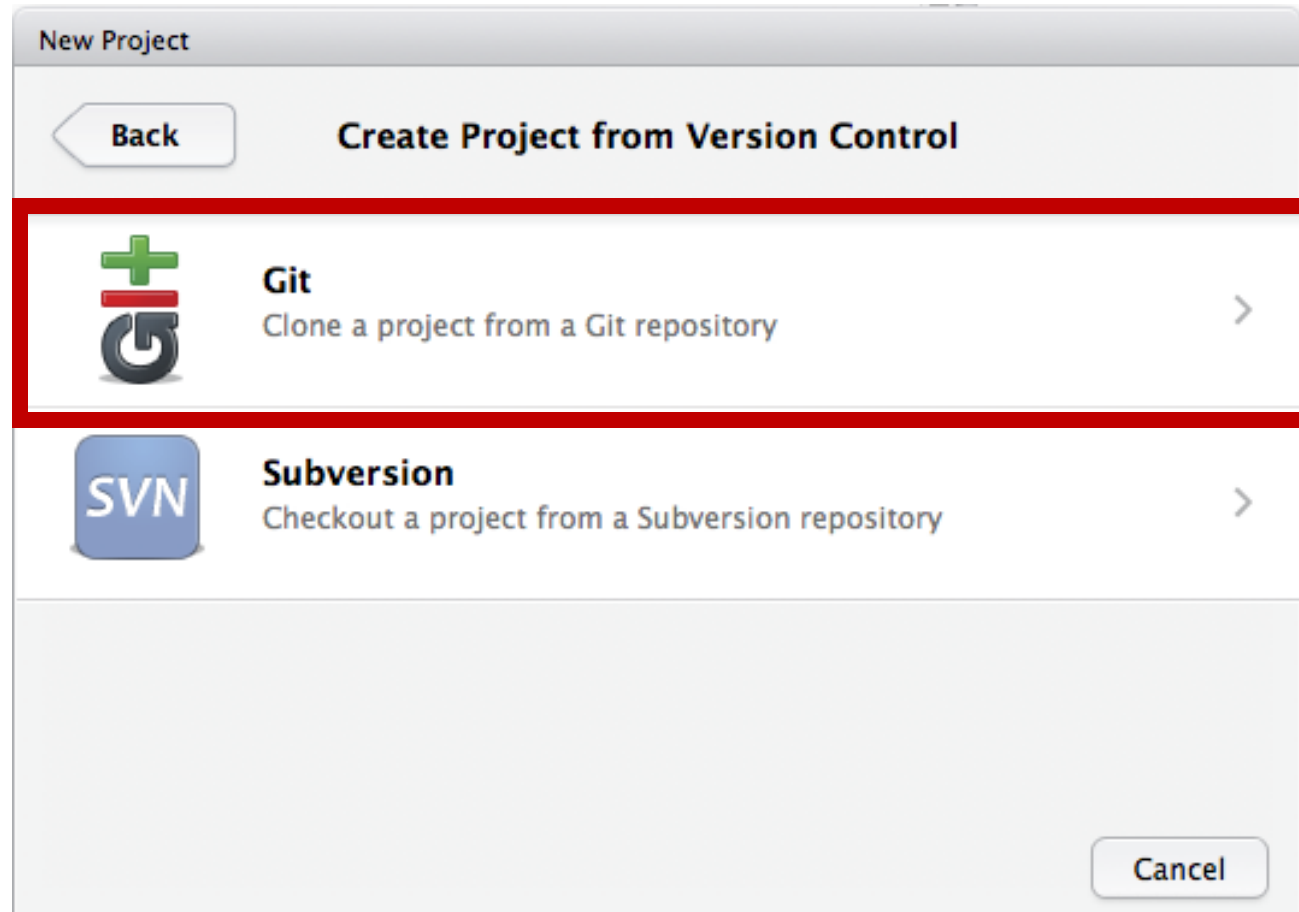
- Select the **File** menu
- Select **New Project**
- A new window will open
- Select **Version Control**



RStudio: Make a New Project

To create a new RStudio Project

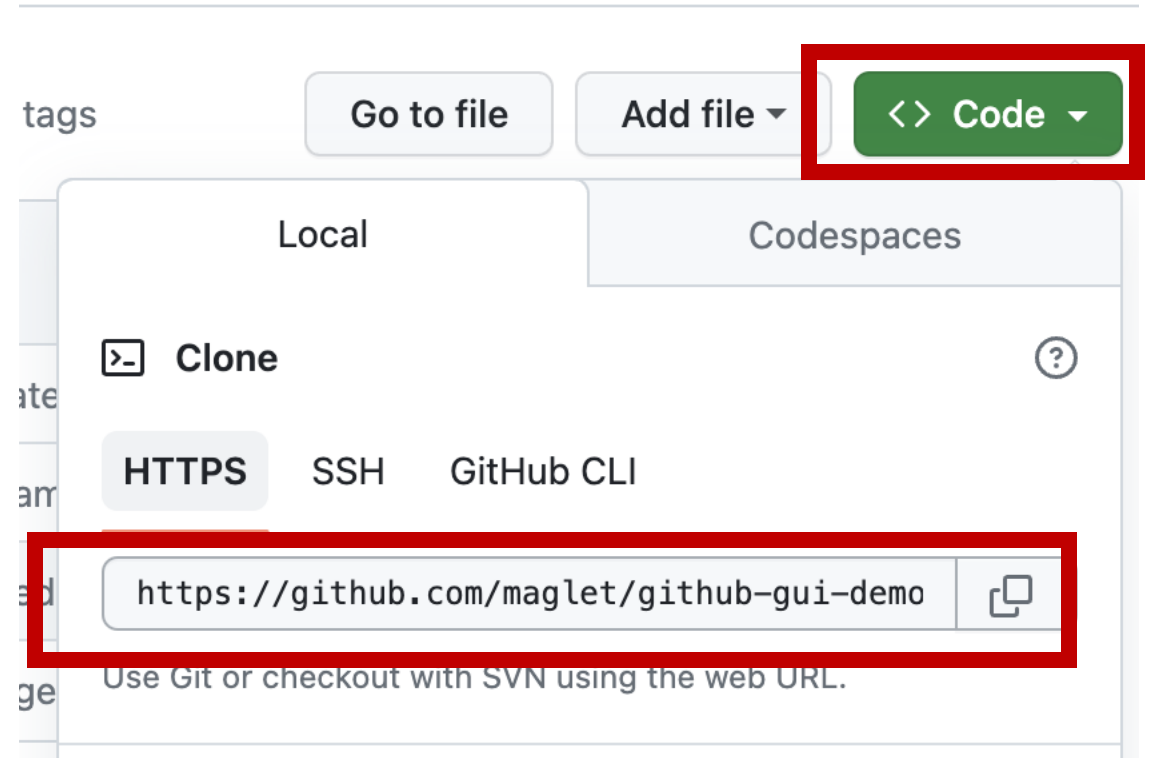
- Select the **File** menu
- Select **New Project**
- A new window will open
- Select **Version Control**
- Select **Git**



RStudio: Make a New Project

To create a new RStudio Project

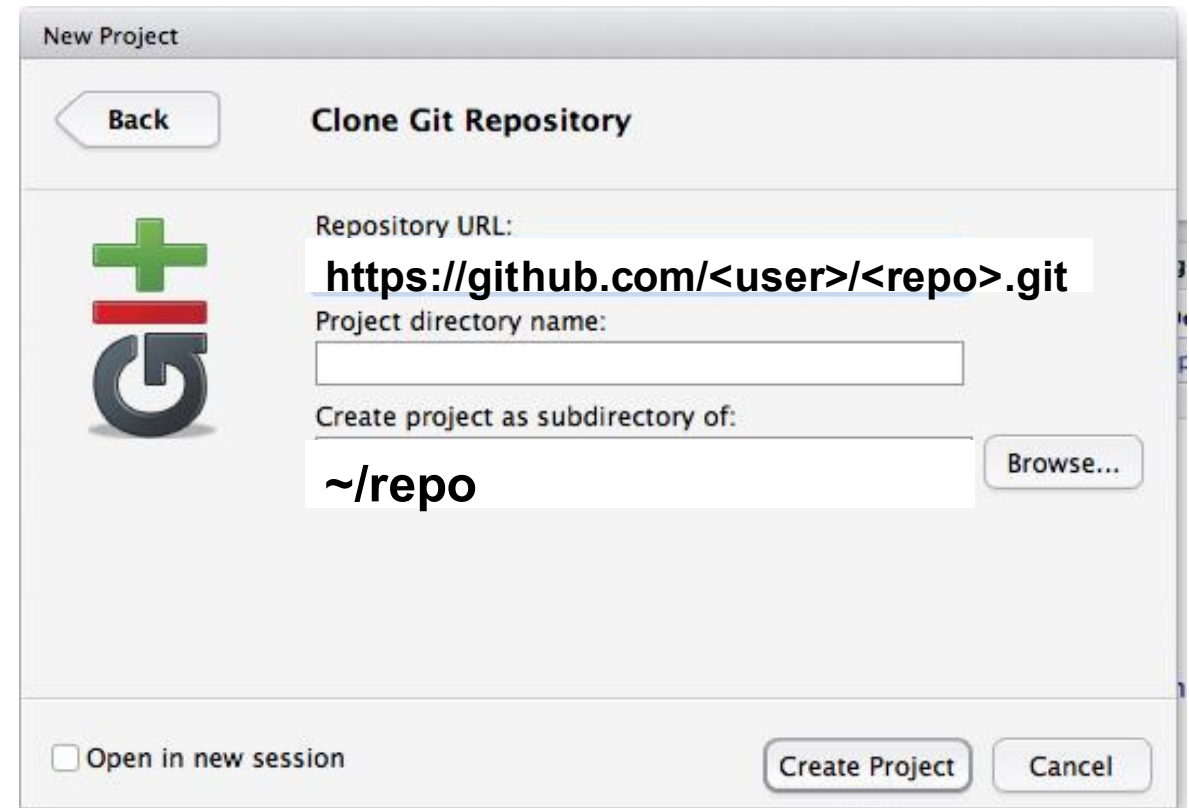
- Select the **File** menu
- Select **New Project**
- A new window will open
- Select **Version Control**
- Select **Git**
- Copy link from GitHub



RStudio: Make a New Project

To create a new RStudio Project

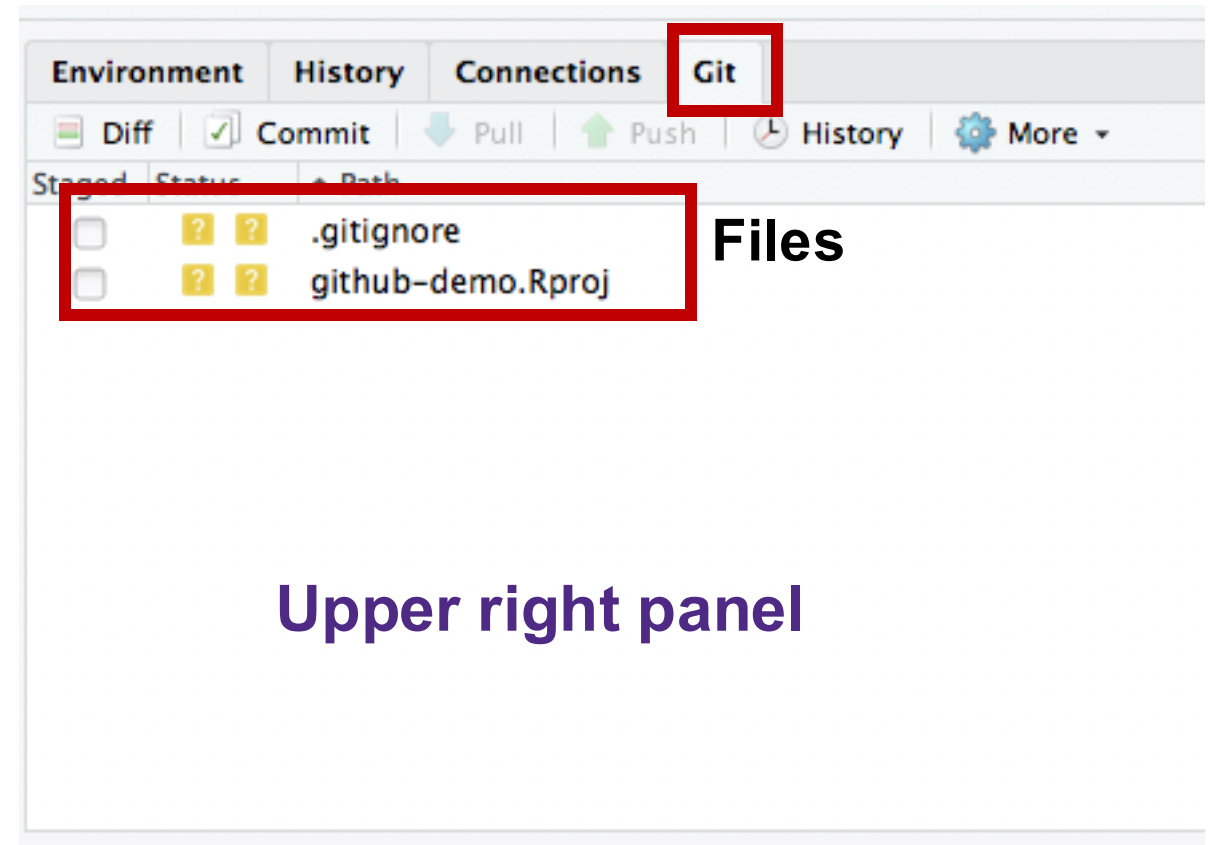
- Select the **File** menu
- Select **New Project**
- A new window will open
- Select **Version Control**
- Select **Git**
- Copy link from GitHub
- Paste into RStudio

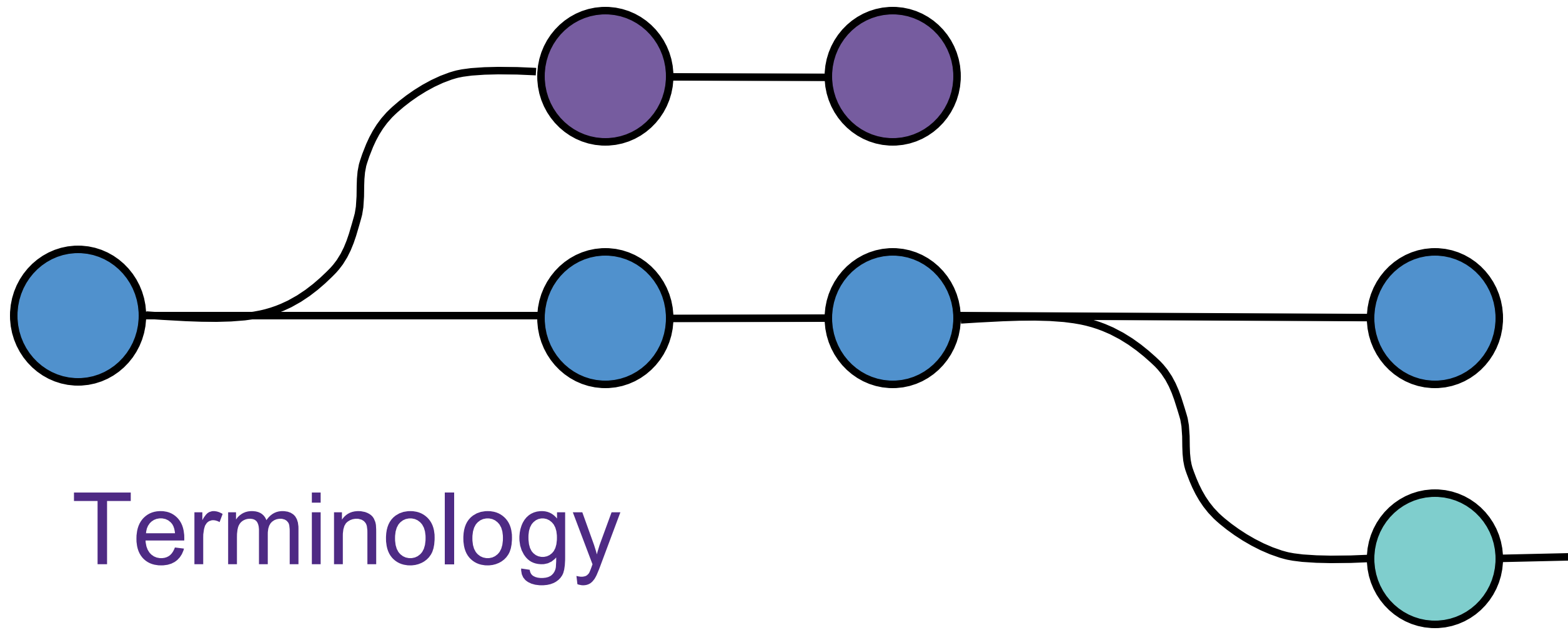


RStudio: Check for git tab

Result: Check for git tab

- A new tab will appear in RStudio: Git
- It will show a subset of files in the repository folder
- We'll explain the jargon next (Stage/Add, commit, etc)





Terminology

Untracked

Edited files recognized by git,
not acted upon

A yellow square containing a white question mark, positioned at the bottom right of the Untracked quadrant.

Added

Edited files to be committed to
your repository

A bright green square containing a white capital letter 'A', positioned at the bottom right of the Added quadrant.

Ignored

Files you don't want to
go in your repository

(Not seen in git tab)

Committed

Files added to your repository

(Not seen in git tab)

Untracked

Add / stage

Added

Edited files recognized by git,
not acted upon

Edited files to be committed to
your repository

ignore

commit

Ignored

Committed

Files you don't want to
go in your repository

Files added to your repository



Untracked



.Rproj



.gitignore

(created by RStudio/git)

?

Added

A

Ignored

Committed

README.md

Untracked



.Rproj **.git**

(created by RStudio)



Added

Goal

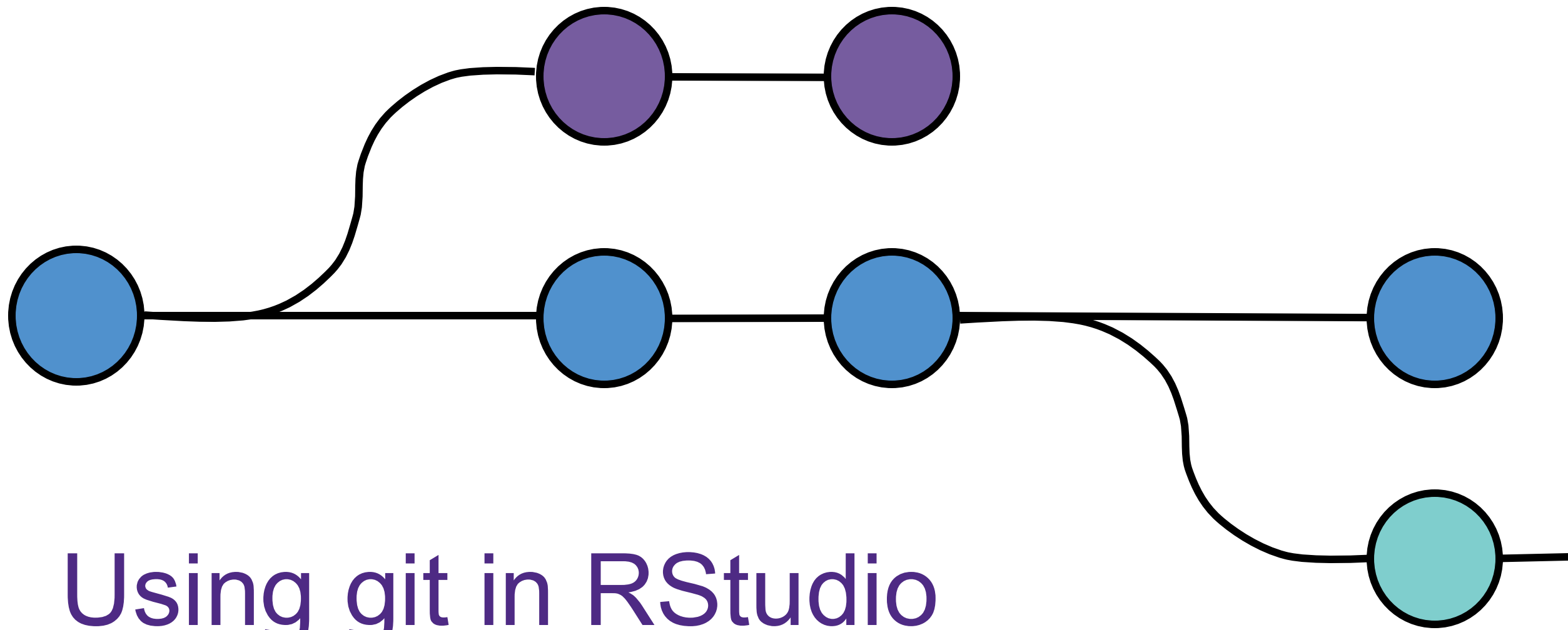
Commit all changes to the repository
(See nothing in the git tab)

Ignored

Staged



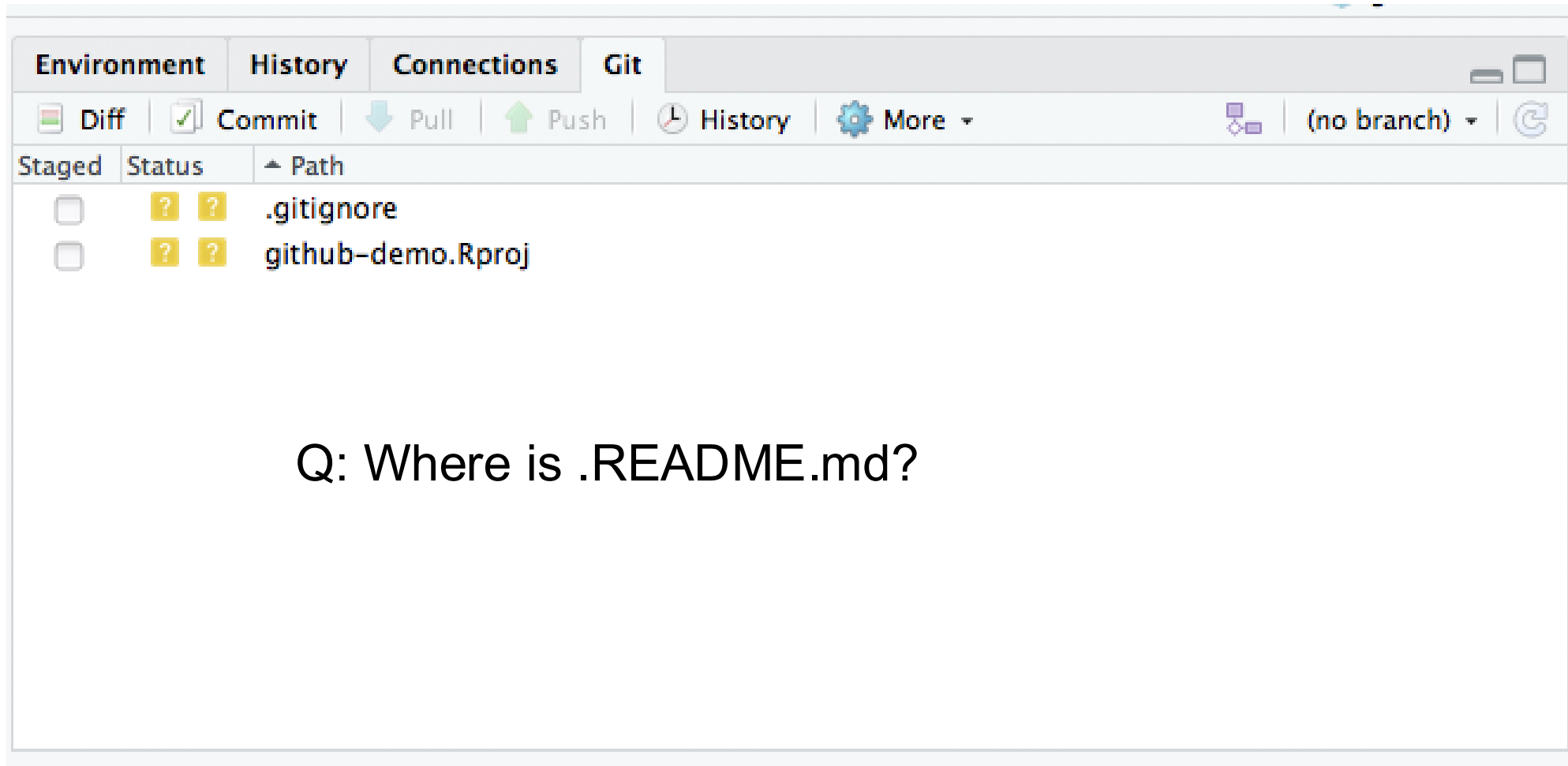
README.md



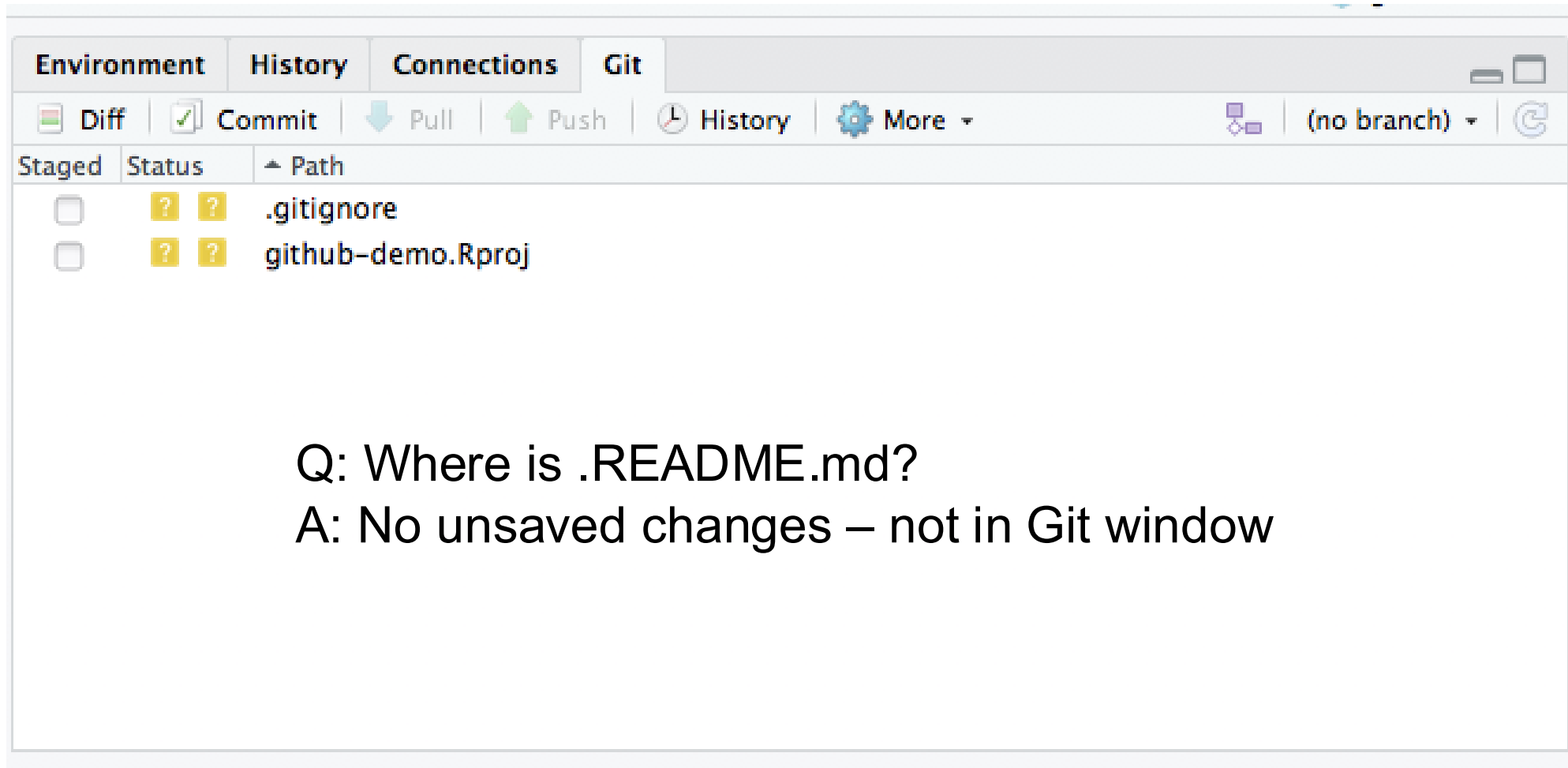
Using git in RStudio

Files in repository folder

Files with uncommitted changes



Files with uncommitted changes



Untracked



.Rproj



.gitignore

?

Added

A

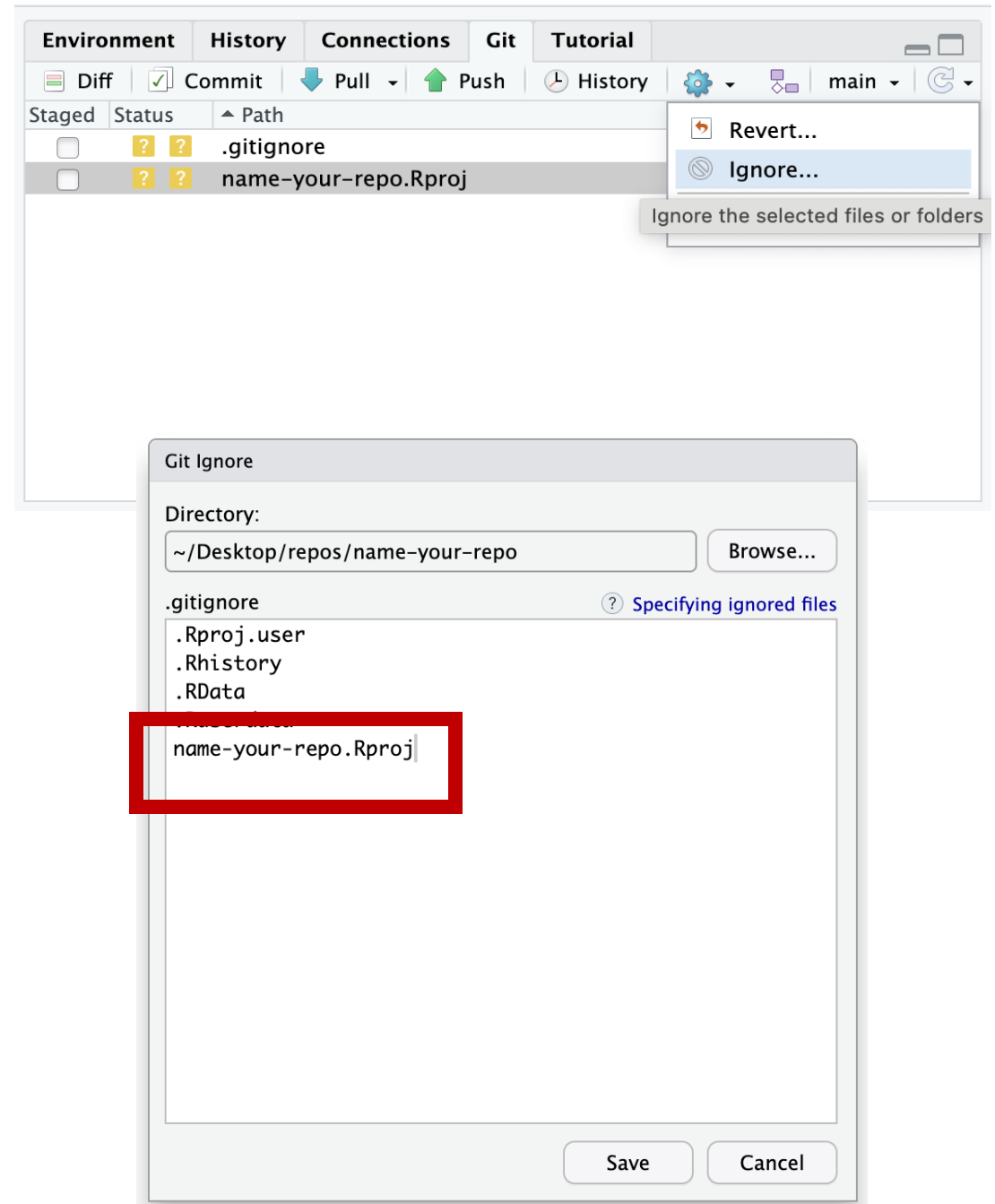
Ignored

Committed

README.md

Ignoring files

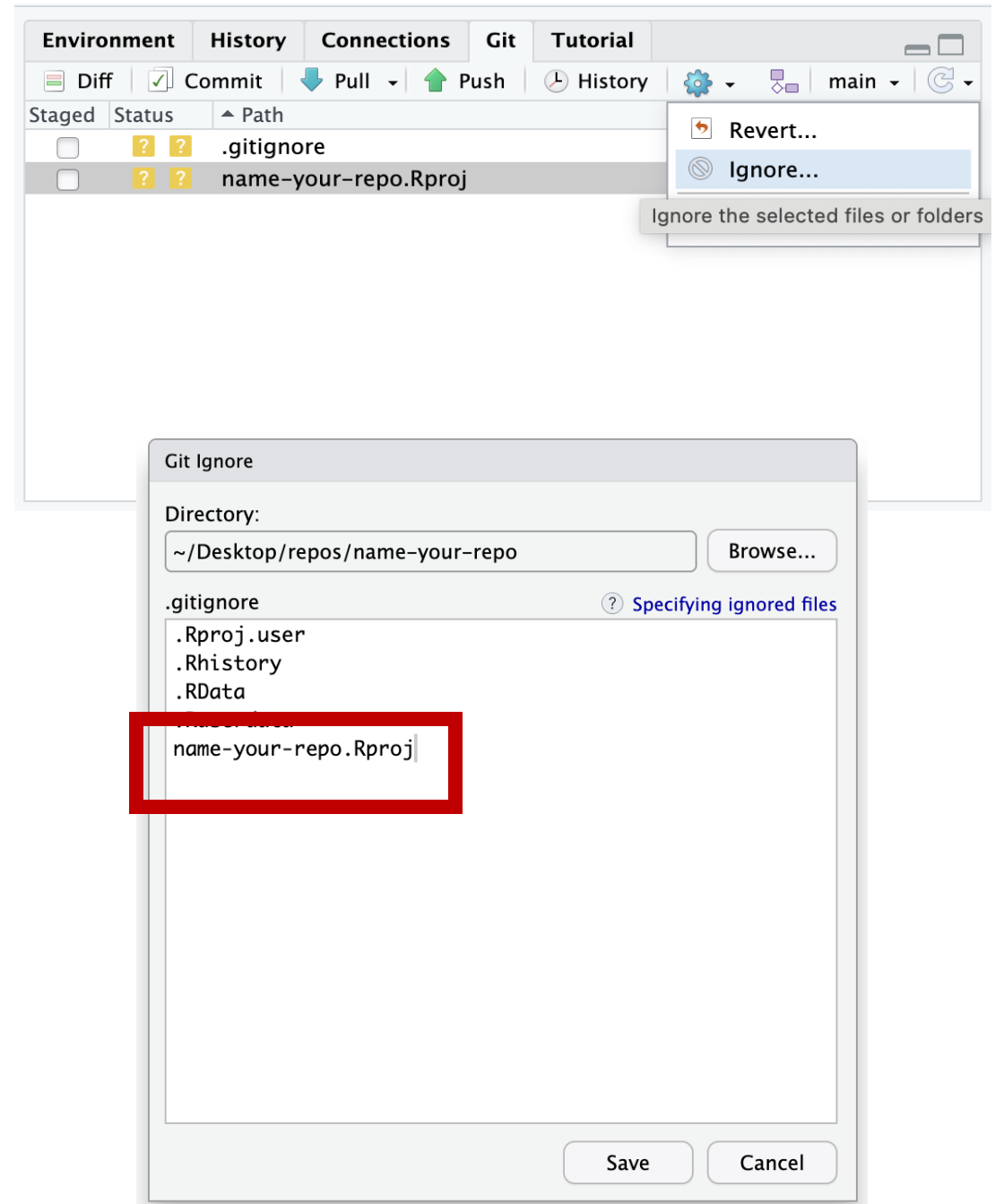
- Git allows you to ignore files that you don't want to track
- .Rproj files are a good example, as it's specific to the filesystem on your computer
- Added the R.proj file to the ignore list



Ignoring files

To ignore a file:

1. Highlight the file you want to ignore
2. Click the gear in the git tab
3. Click ignore – the file name will appear in the ignore list
4. Click Save



Untracked



.Rproj



.gitignore

ignore

?

Ignored

Added

A

Committed

README.md

Untracked



.gitignore

?

Added

A

Ignored



.Rproj

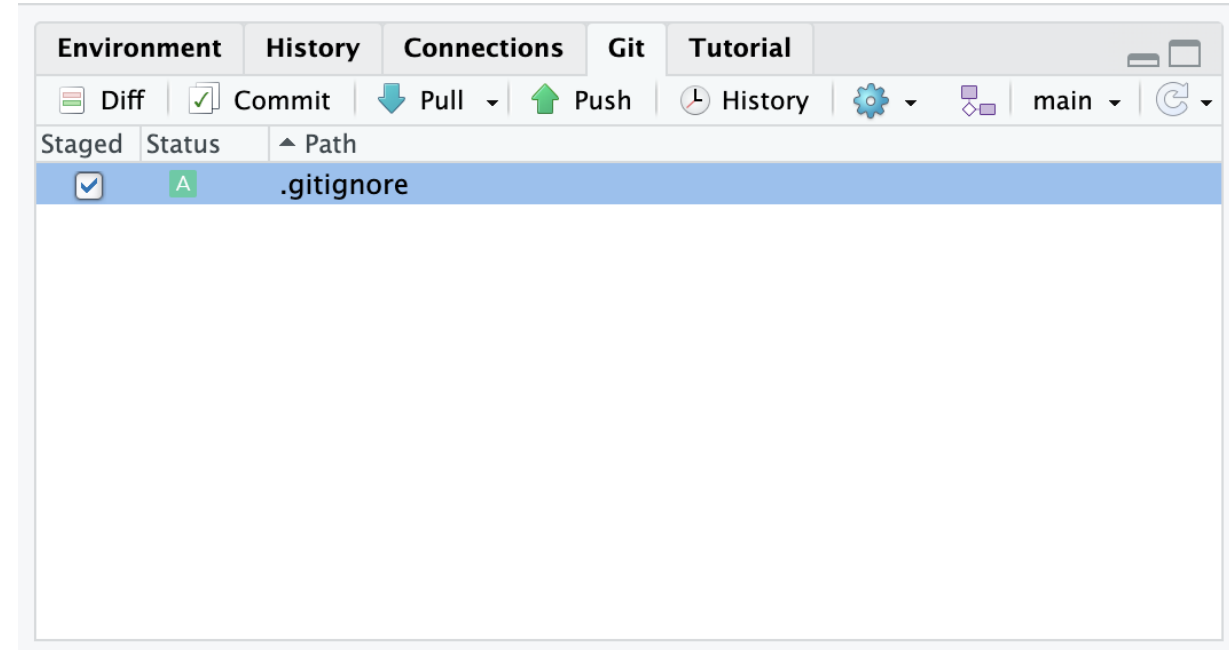
Committed

README.md

Adding/Staging files

The first step to saving your changes to your repo is to **Add/Stage** the file

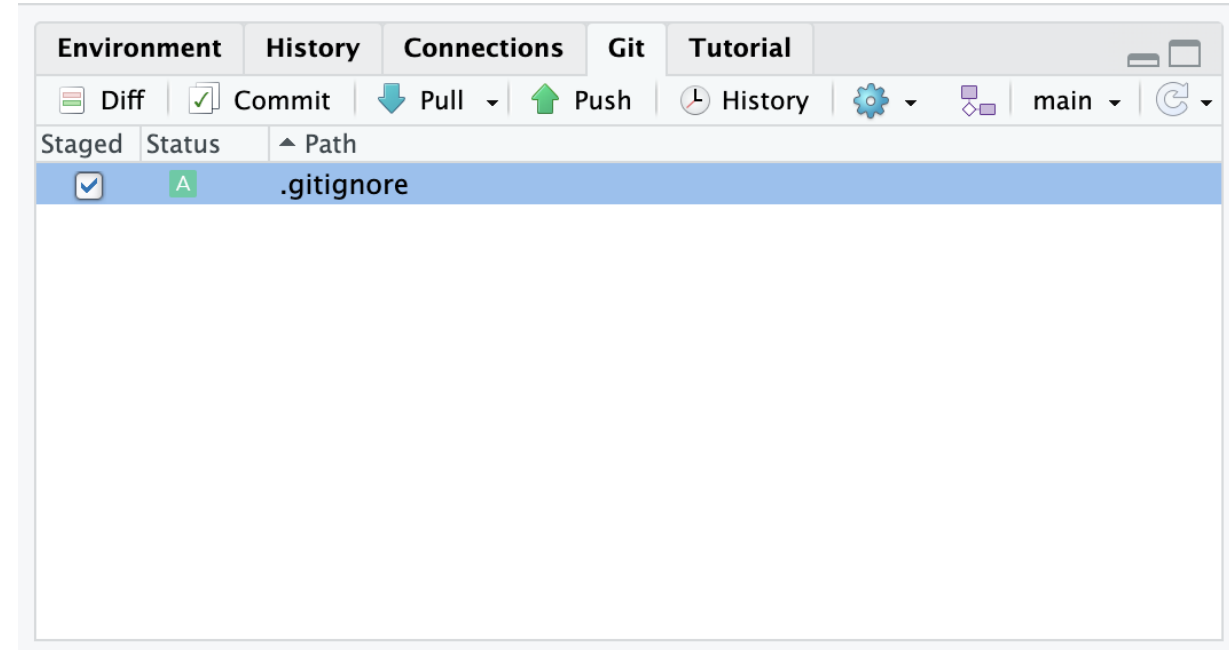
- Add is git terminology
- Stage is RStudio terminology
- Adds files to a "Staging area"
- Allows you to combine changes to multiple files into one "Save"



Adding/Staging files

To stage a file:

1. Check boxes to add files to the “staging area”
2. **Status:** **A** for Added
3. Staging area compiles changes to be committed to the repository



Untracked

Add / stage

Added



.gitignore

?

A

Ignored

Committed



.Rproj

README.md

Untracked

Added



.gitignore

?

A

Ignored



.Rproj

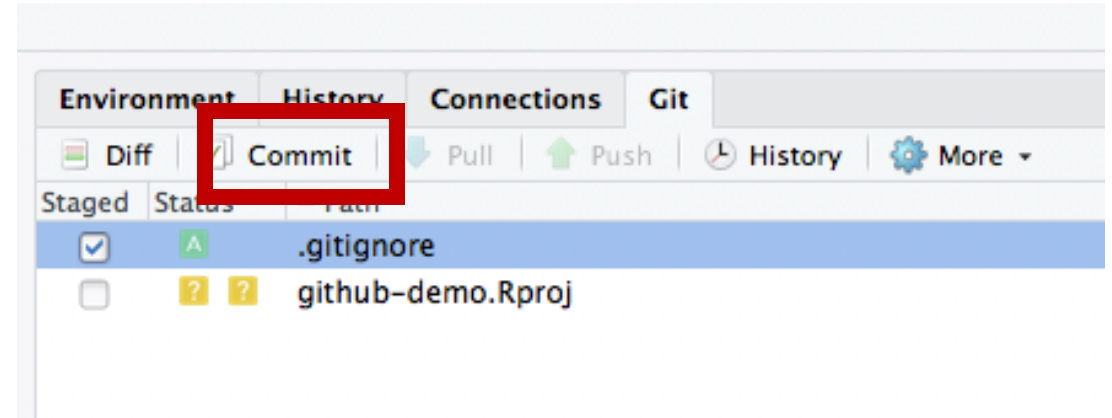
Committed

README.md

Committing files

After adding files to the staging area, you must **commit** them to save them to your repo

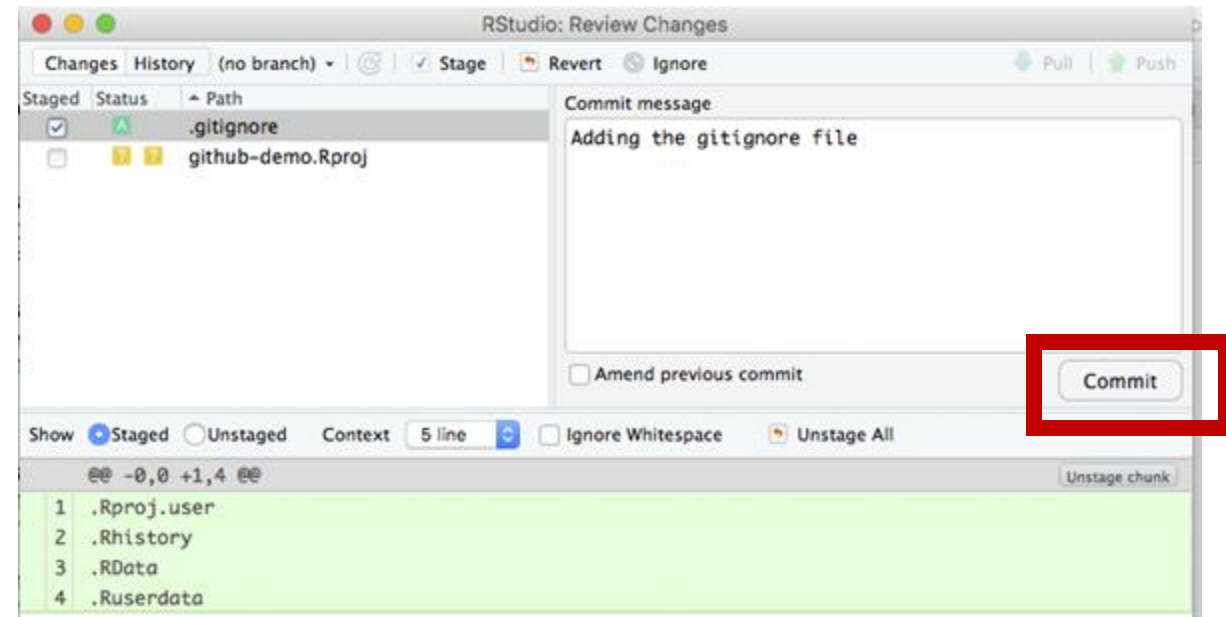
- The **commit** action adds all files from the staging area to the repository
- Git will ask you for a **commit message** that describes what you changed in this commit



Committing files

You will also be able to see what has changed by looking at the bottom of the screen

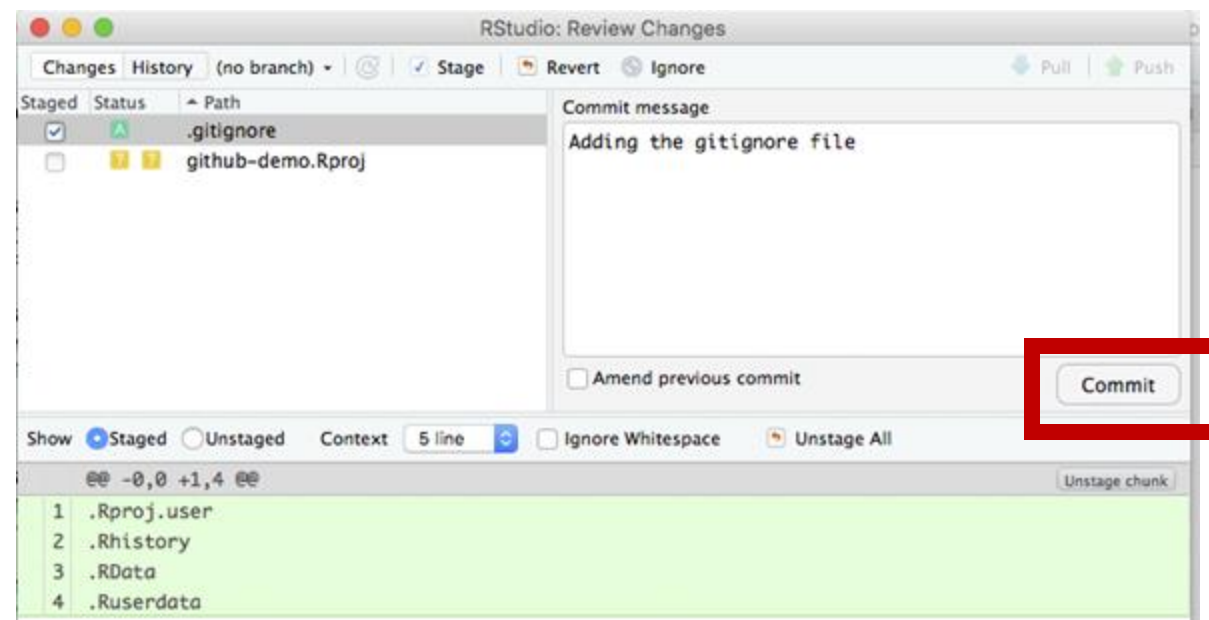
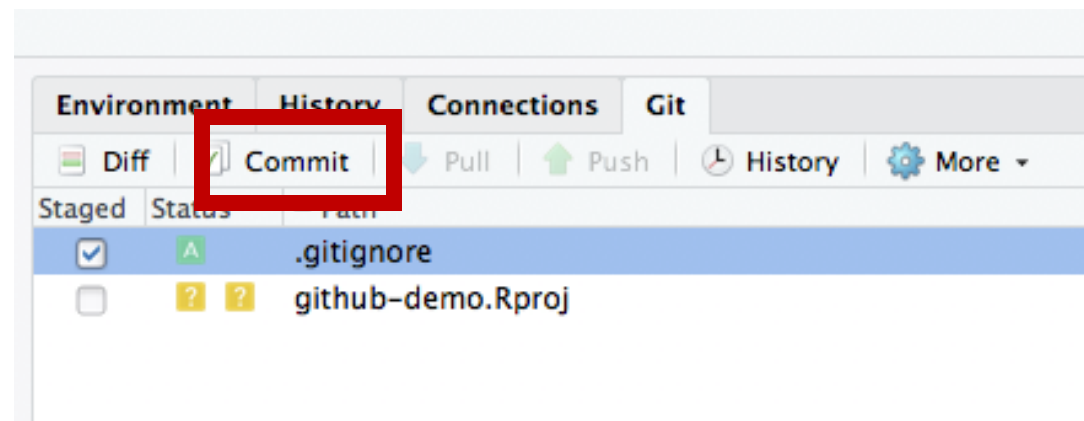
- Text that was added will be in Green
- Text that was removed will be in Red



Committing files

To commit your file(s):

1. Click the **Commit** button in the Git tab
2. Type a description of your changes in the **commit message** box
3. Click **Commit**



Untracked

Added



.gitignore

?

commit

A

Ignored



.Rhistory



.Rproj

Committed

README.md

Untracked

Added

?

A

Ignored

Committed



.Rhistory



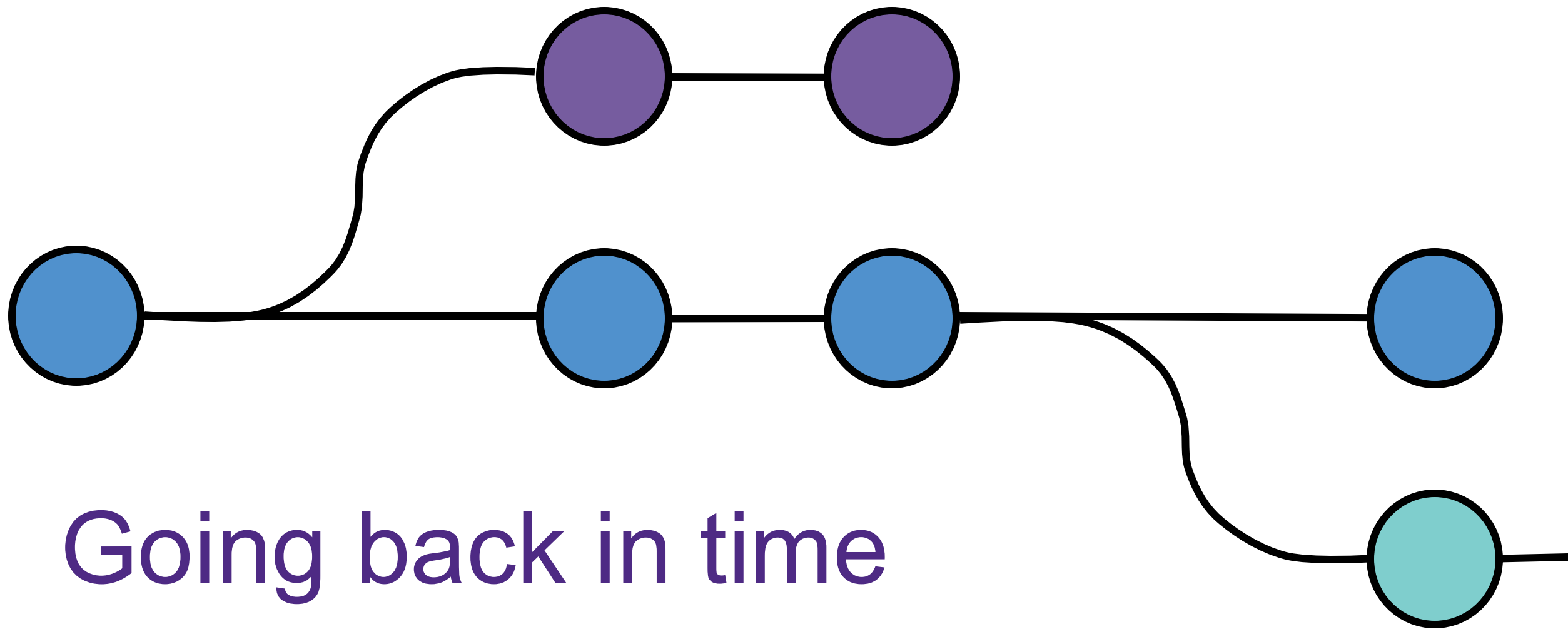
.Rproj

.gitignore

README.md

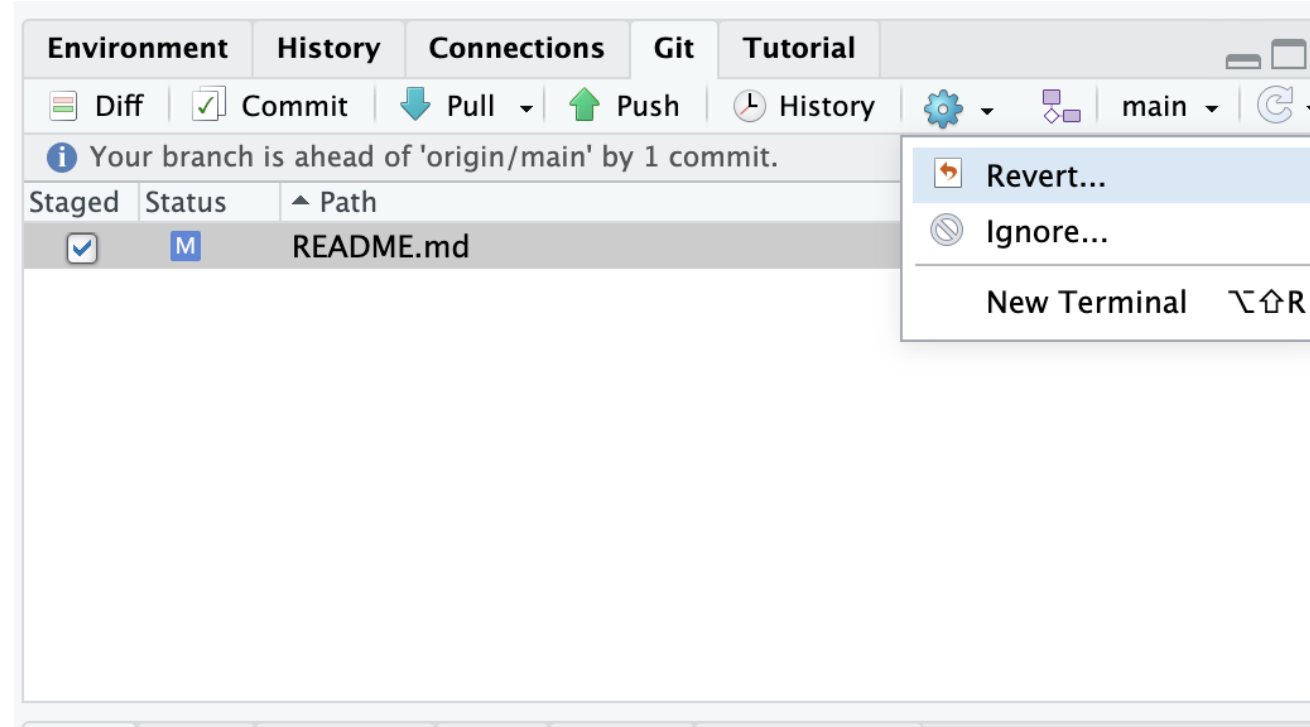
Exercise:

1. Edit your README.md file
2. **Stage changes: Add** the changes file to the **staging area**
3. **Commit** your staging area to the **repository**



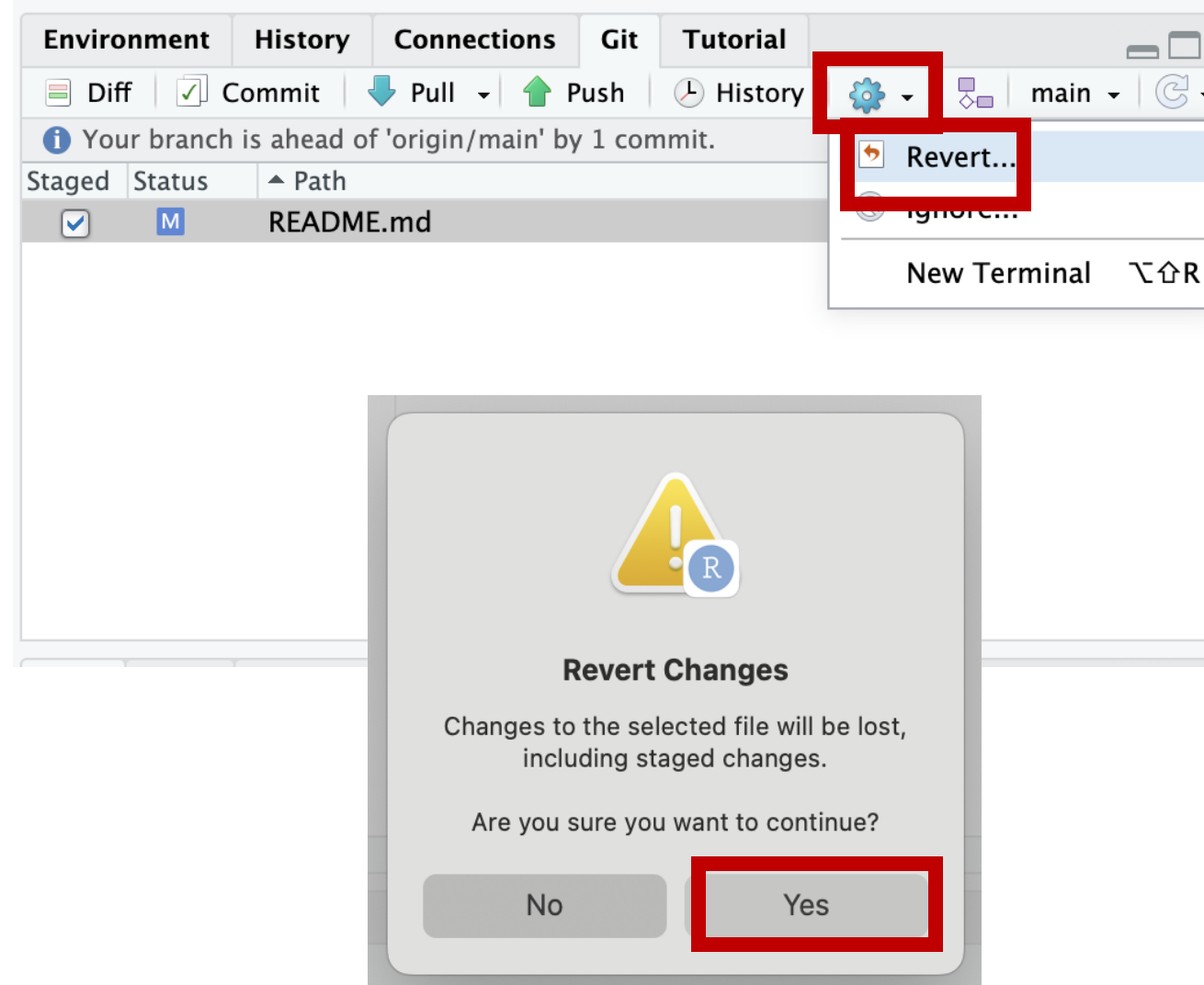
Reverting changes

- Removes all uncommitted changes
 - Unstaged changes
 - Staged changes
- Can't go back to previous commits within the from R Studio IDE



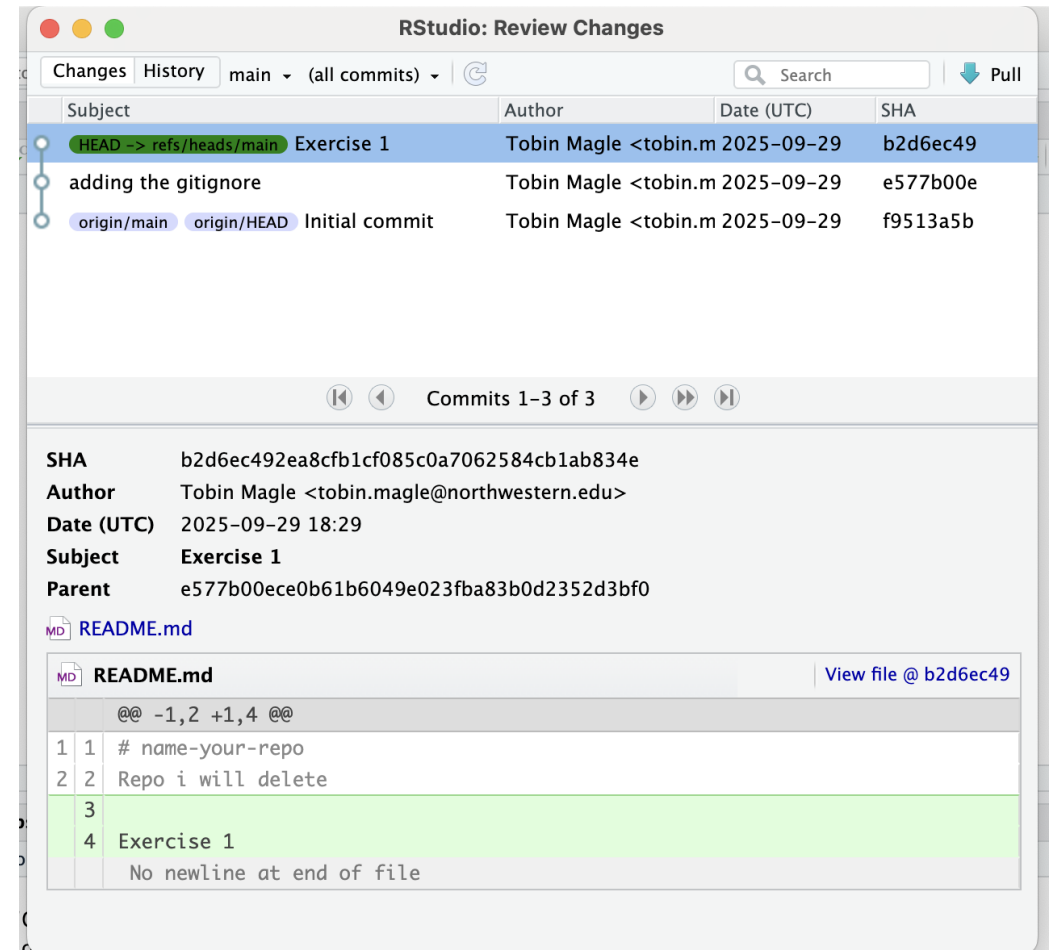
Demo: what can you revert?

1. Add some lines to your README.md file
2. Don't add or commit them > Revert
3. Add them, but don't commit them > Revert



Viewing previous commits

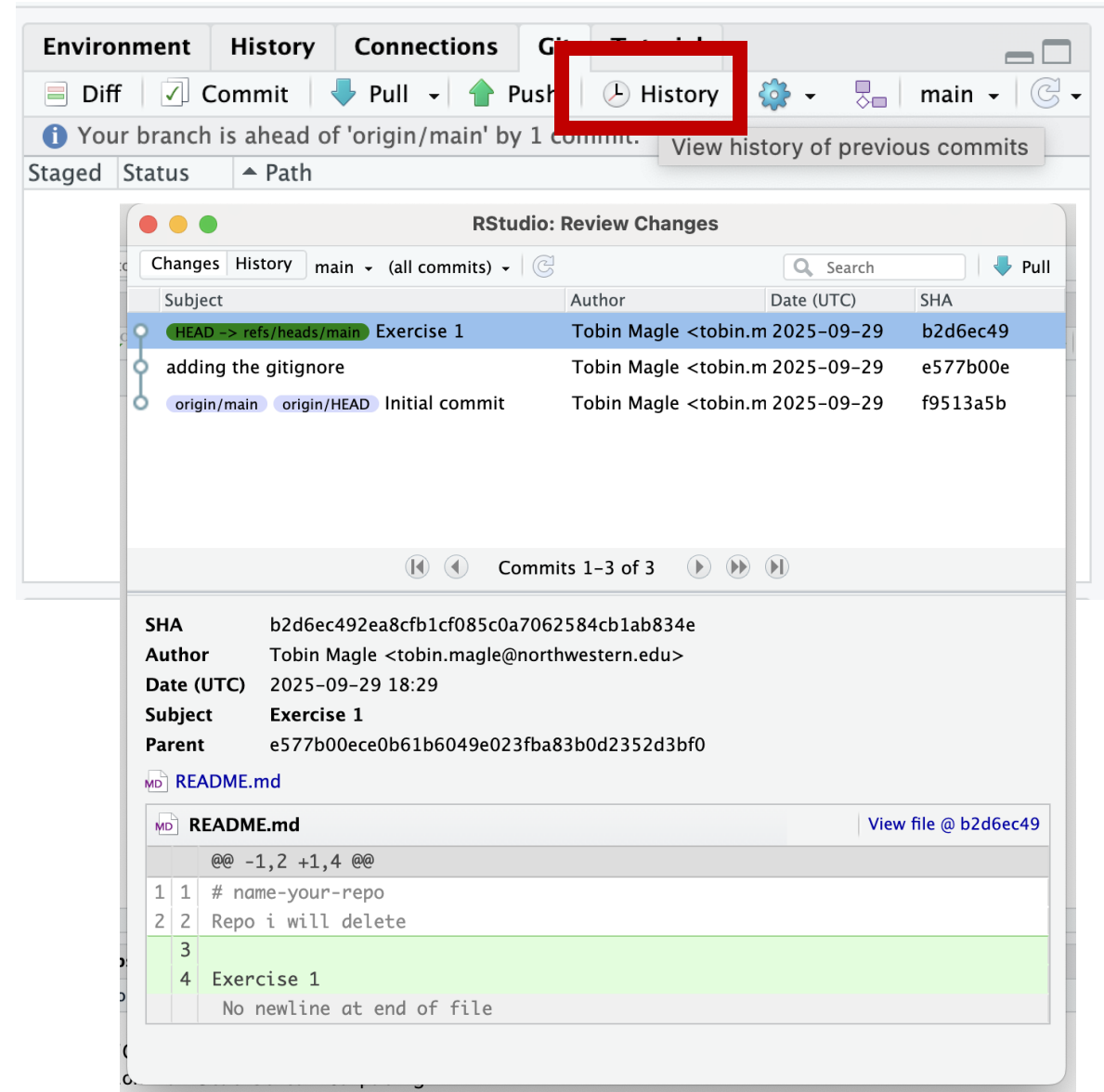
- The **History** window allows you to see previous commits
- One line for each commit
 - Commit message
 - Who made the commit
 - When they made it
 - SHA (aka hash) - unique identifier for the commit
- Click on the commit to see what changes were made



Viewing previous commits

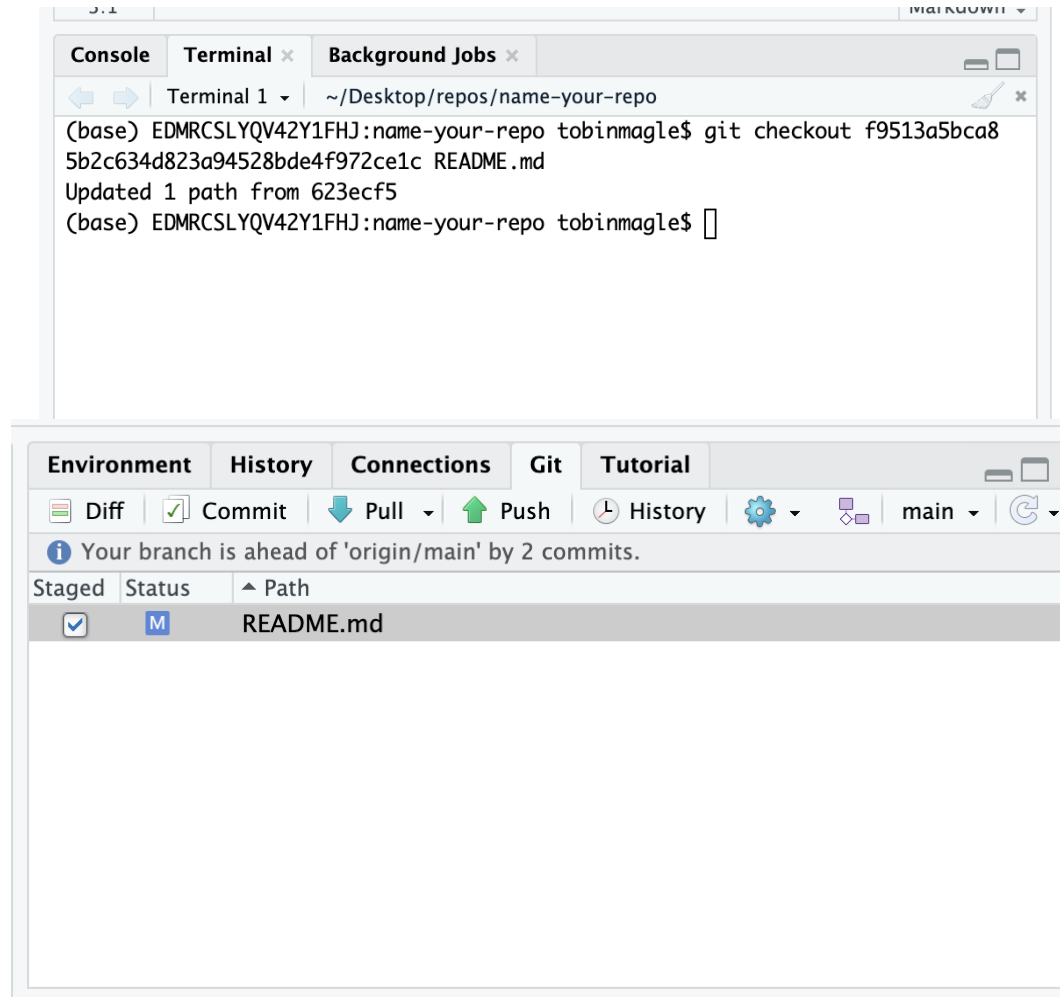
To view previous commits:

1. Click on "History" in the Git tab
2. A new window will open
3. Click the commit you would like to see details about



Check out a previous version

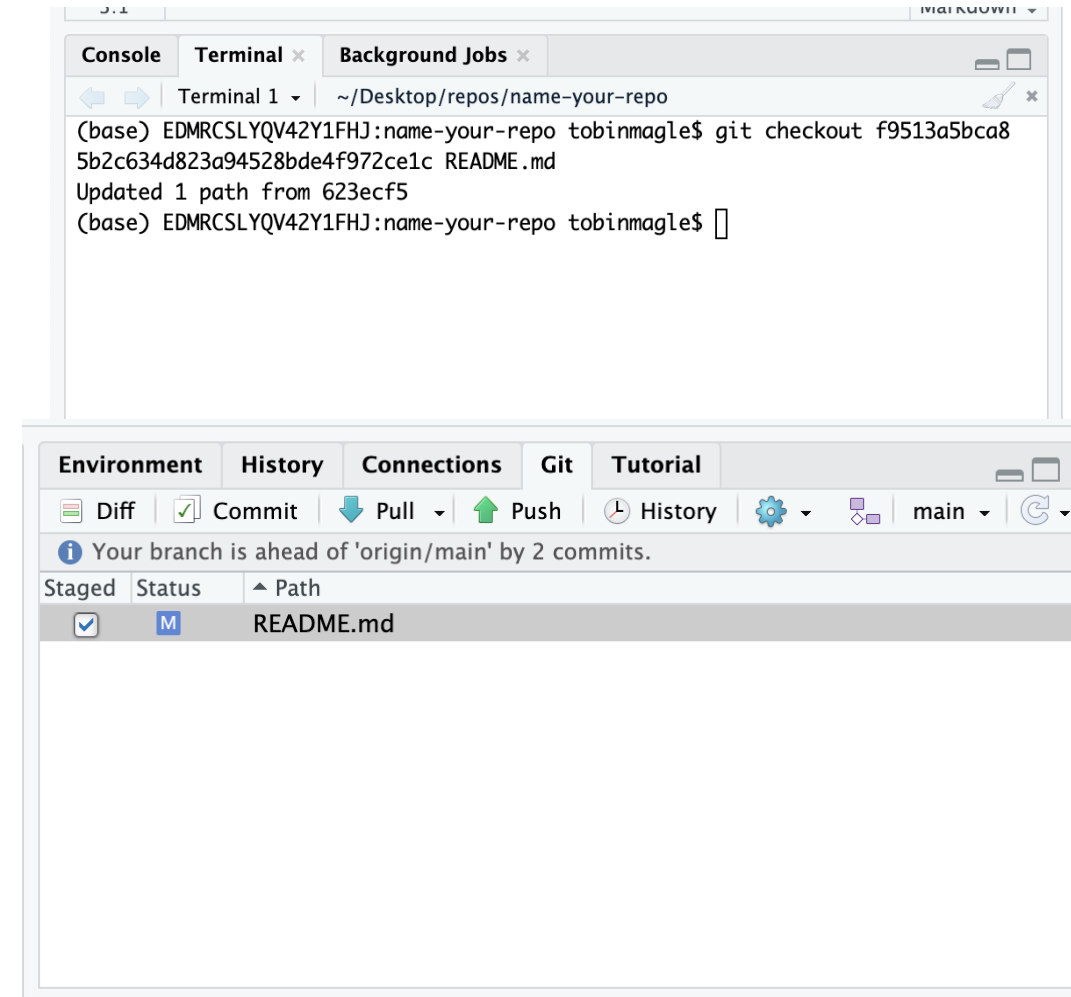
- Checking out a previous version pulls an old version of a file and puts it in your staging area
- File will show as modified, add and commit like normal
- Need to use the terminal

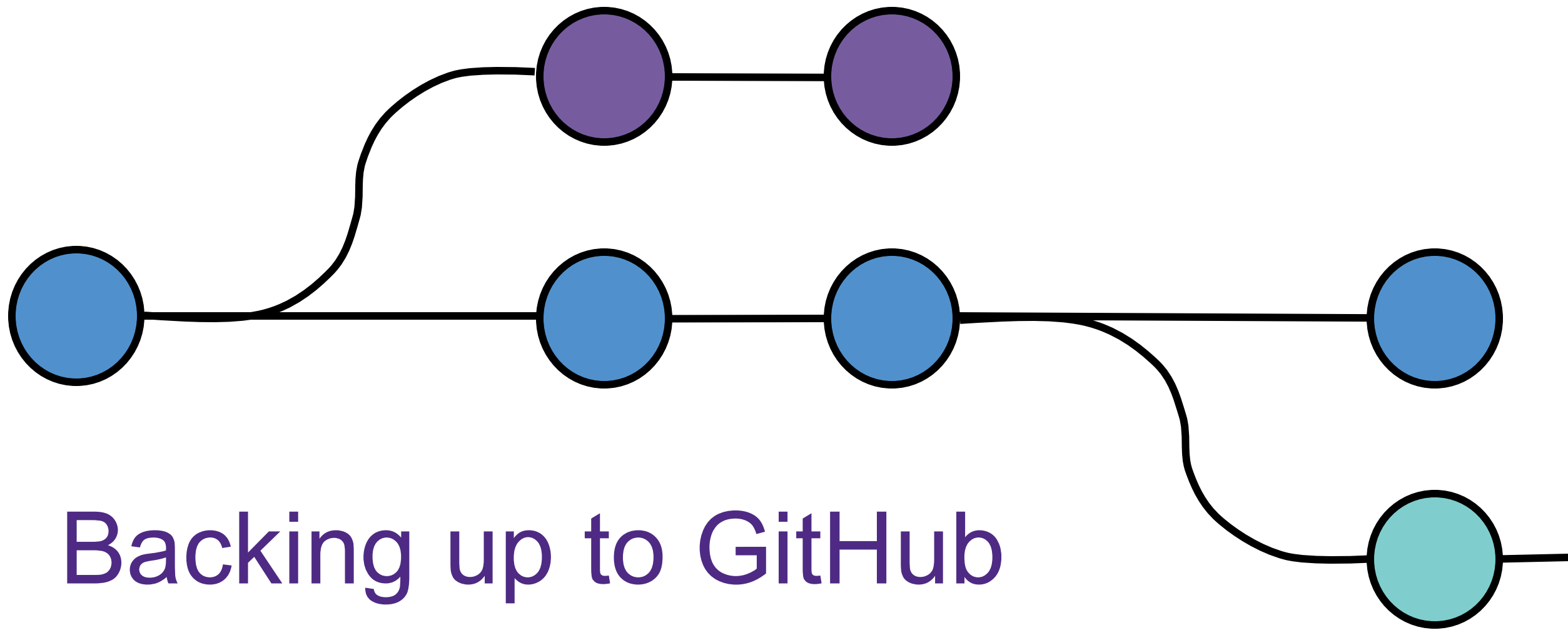


Check out a previous version

To check out a file from a commit:

1. Copy the SHA hash from the History window
2. Open the terminal in RStudio
3. Run the following command
`git checkout <SHA hash> <filename>`
4. File will show as modified in the file window
5. Add and commit if you want to keep the changes

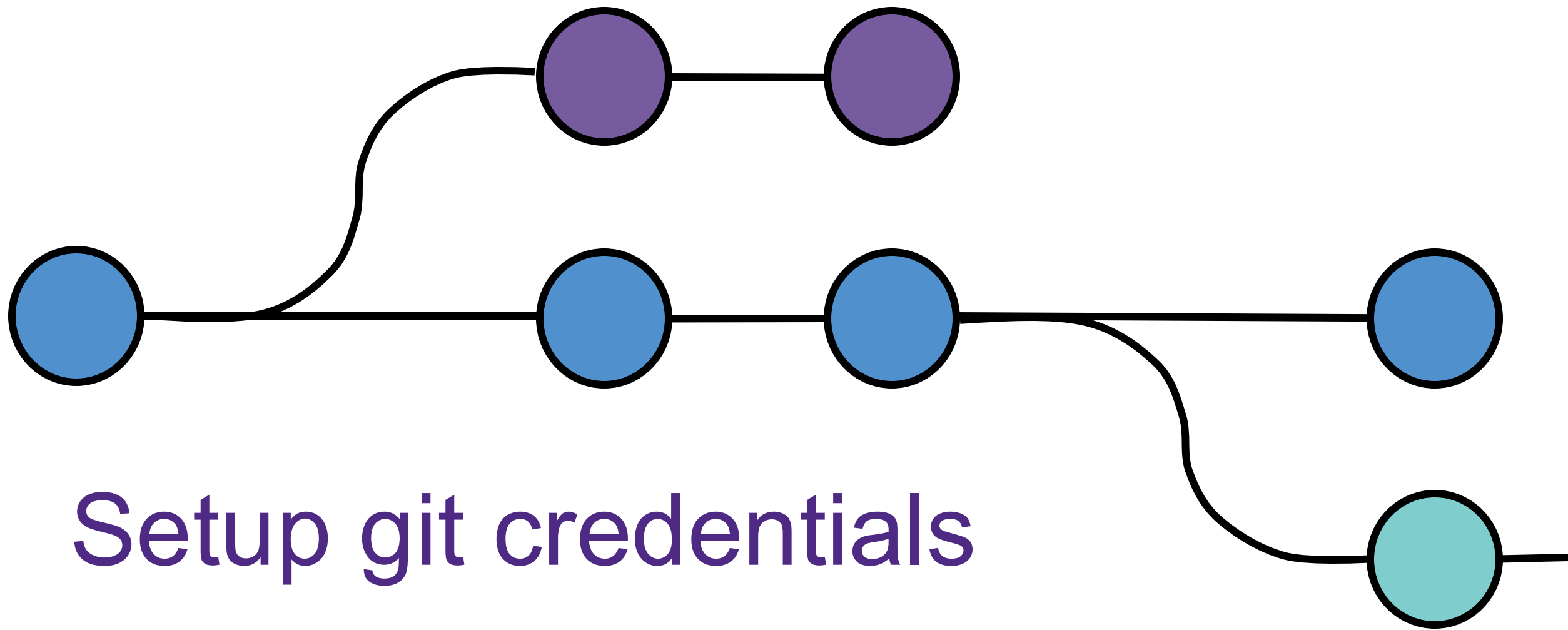




GitHub best practices

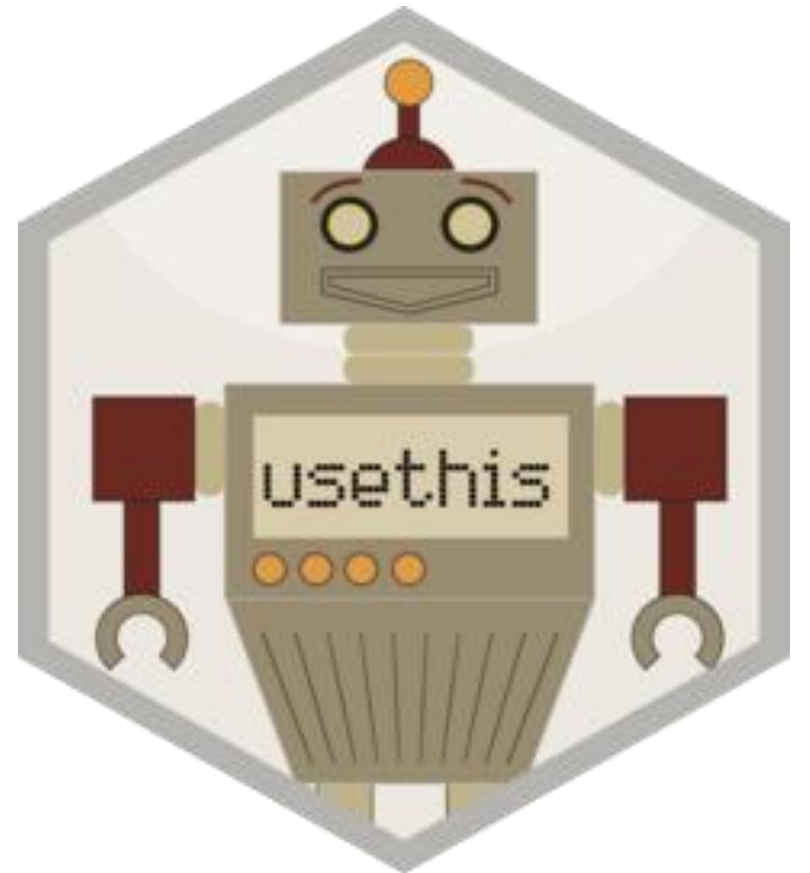
- Store **code and documentation**, not data
- Never store **protected or personal data** even in a private repo
- Store **credentials** (private keys or tokens) in a separate file outside of your repository
 - ie, not in your code
- Add a **personal email** to your GitHub account, so you can use it when you leave Northwestern





Configure git on your computer

- Authentication in GitHub has gotten complicated lately
- The `usethis` package "automates repetitive tasks that arise during project setup and development"
- We will use this package to
 - Configure your git user on your computer
 - Set up access credentials in GitHub



Configure your git user on your computer

1. Open RStudio console

In RStudio console

2. Install and load `usethis`

```
install.packages("usethis")  
library(usethis)
```

3. Set git username and email

- Commit history shows username
- **Use your GitHub account email**

```
usethis::use_git_config(user.name="Jane Doe",  
user.email="jane@example.org")
```

4. Set GitHub authentication

```
usethis::create_github_token()
```

Authenticating into GitHub

Step 1: Generate a GitHub token

- Good for security
- Use the token instead of your password
- Access tokens allow you to give different levels of access than you have as a user
- Tokens expire
- You can revoke access to tokens



Generate a GitHub token

To generate a GitHub token

1. Run `create_github_token()`
2. A webpage will open: New personal access token (classic)
3. Options pre-selected to let you push and pull from a GitHub repo
4. Scroll down and click **Generate Token**

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

DESCRIBE THE TOKEN'S USE CASE

What's this token for?

Expiration

📅 30 days (Oct 29, 2025) ▼

The token will expire on the selected date

Select scopes

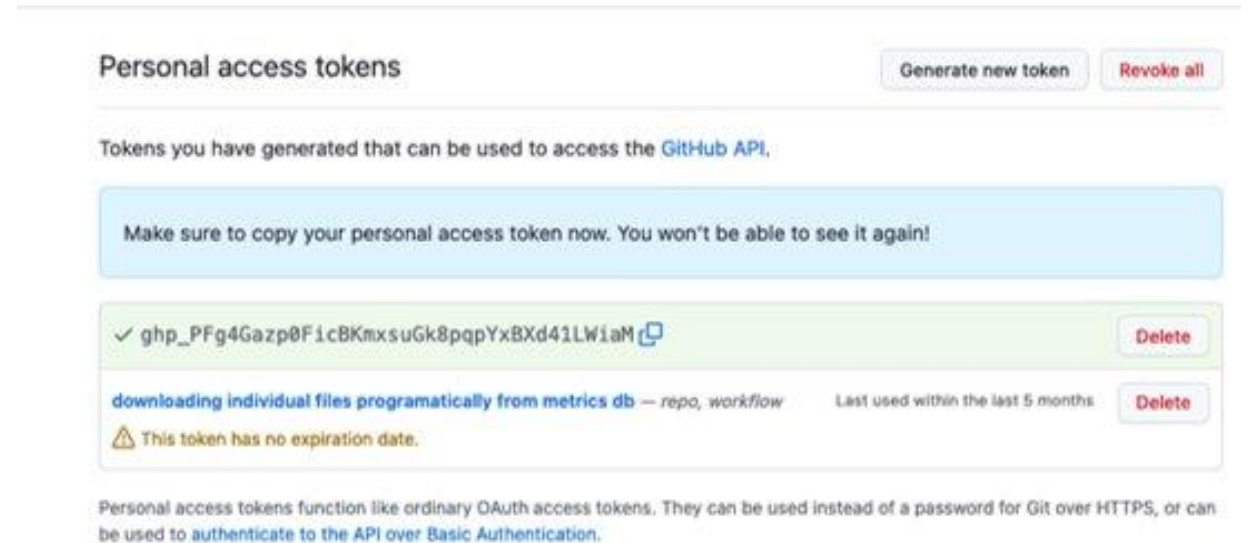
Scopes define the access for personal tokens. [Read more about OAuth scopes](#).

- | | |
|---|--------------------------------------|
| ☒ repo | Full control of private repositories |
| ☒ repo:status | Access commit status |
| ☒ repo_deployment | Access deployment status |
| ☒ public_repo | Access public repositories |
| ☒ repo:invite | Access repository invitations |
| ☒ security_events | Read and write security events |
| ☒ workflow | Update GitHub Action workflows |

Generate token

Generate a GitHub token

1. Run `create_github_token()`
2. A webpage will open: New personal access token (classic)
3. Options pre-selected to let you push and pull from a GitHub repo
4. Scroll down and click **Generate Token**
5. A new page will open with the token. Copy it!



Generate token

Authenticating into GitHub

Step 1: Generate a GitHub token

Step 2: Use the token to connect the repository on your computer to your repository on GitHub

- Use the token from Step 1 to establish the connection



Log in to GitHub

To log in to GitHub:

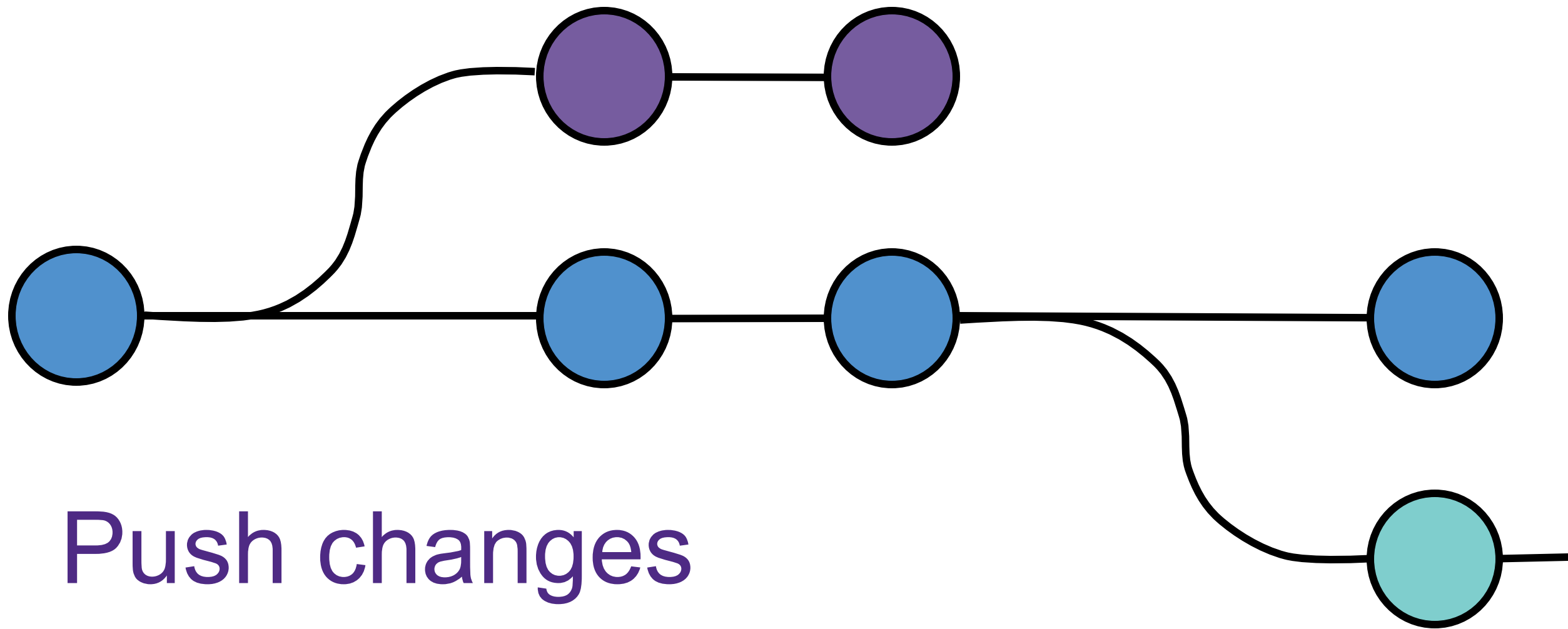
1. Run the `gitcreds_set()` command
2. Paste in the token when asked
3. If you don't have GitHub credentials, it will ask you to for the Token
4. If you had existing credentials you will see 3 options (Pick 2 to use new Token):

```
1: Abort update with error, and keep  
the existing credentials  
2: Replace these credentials  
3: See the password / token
```

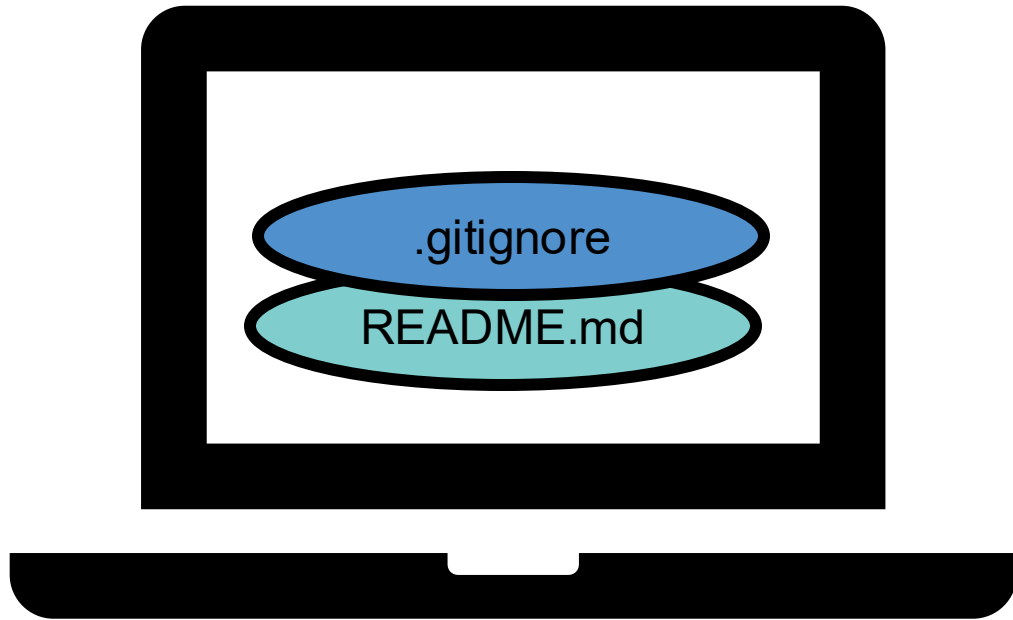
5. Git saves these credentials for you

In RStudio console

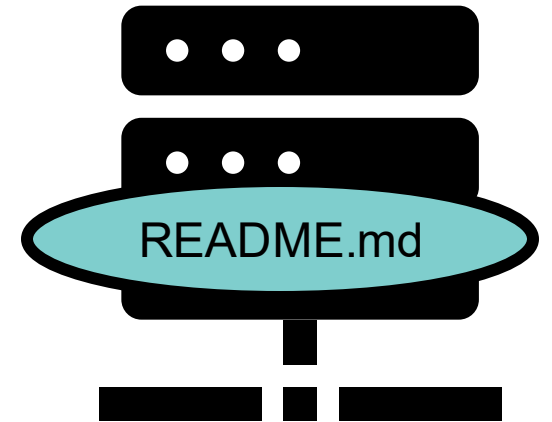
```
gitcreds::gitcreds_set()
```



Local changes not backed up

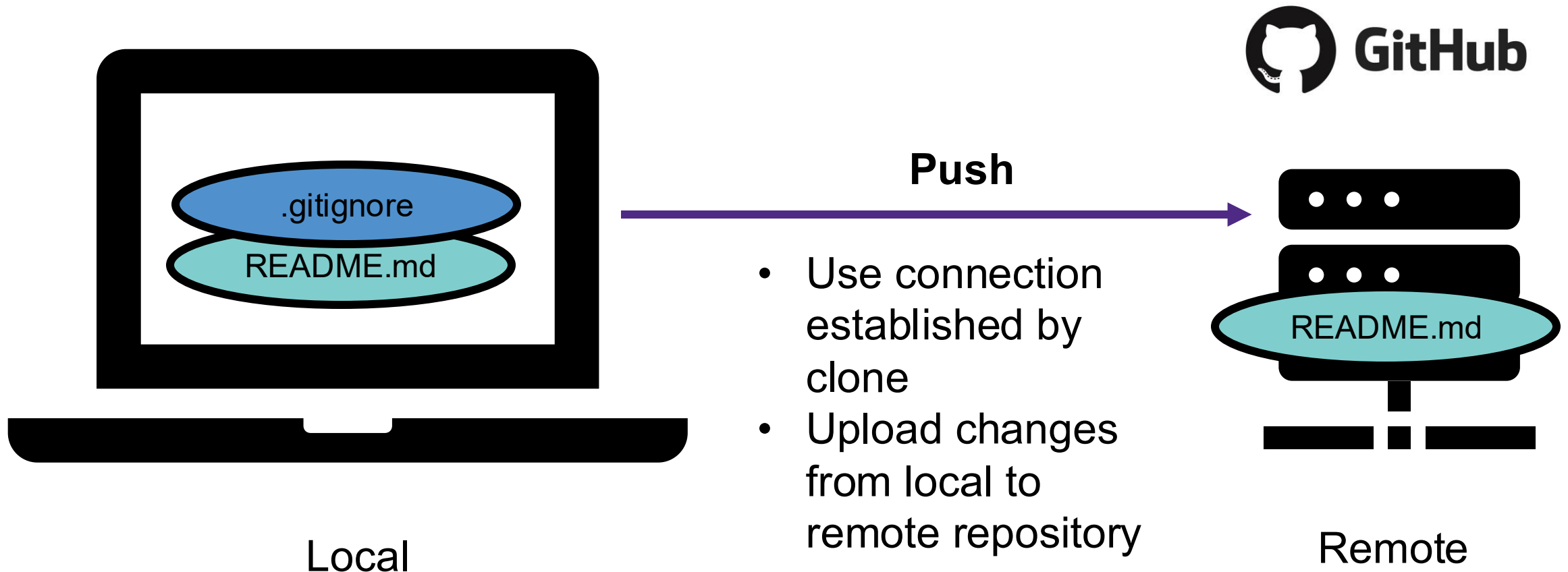


Local



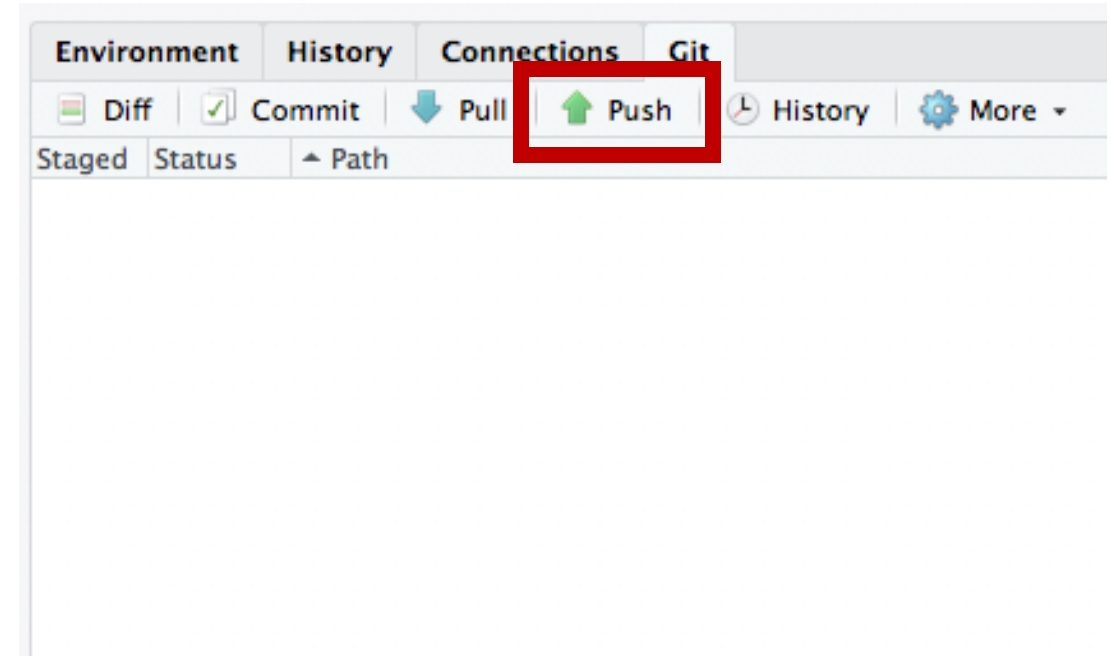
Remote

Push to remote

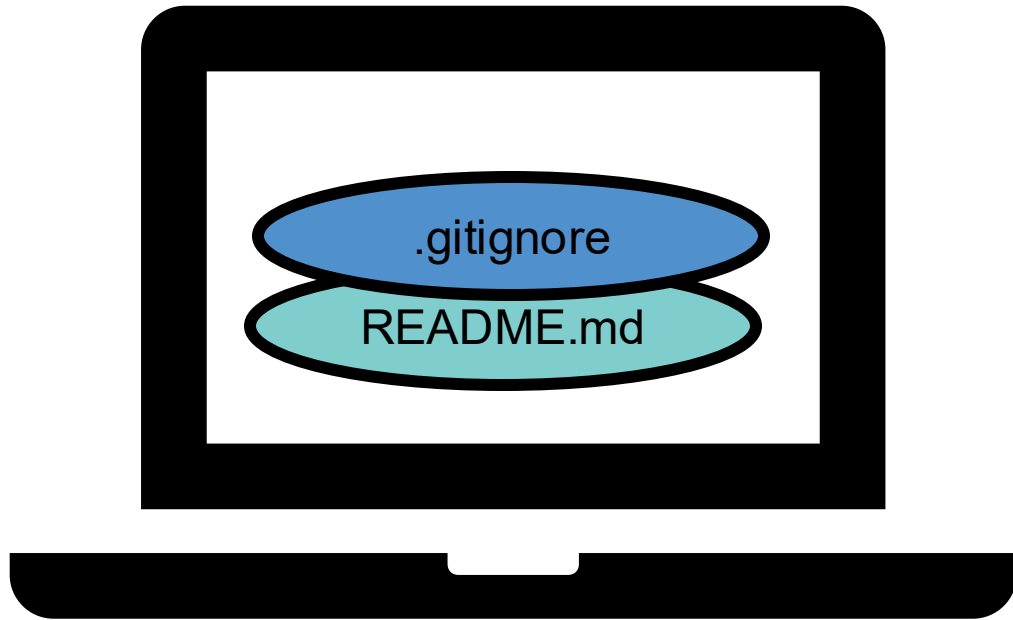


Push

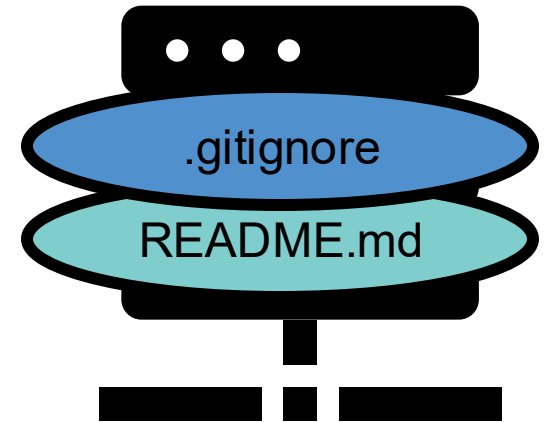
- Save local changes to remote repository
- Changes must be committed before pushing
- Requires that you have authentication set up on your machine



Changes pushed



Local



Remote

Files in remote

 **maglet / github-demo**

Unwatch 1

Star 0

Fork 0

<> Code

Issues 0

Pull requests 0

Projects 0

Wiki

Insights

Settings

No description, website, or topics provided.

[Add topics](#)

2 commits

1 branch

0 releases

0 contributors

Branch: master

New pull request

Create new file

Upload files

Find file

Clone or download

 **Tobin Maple** Added .Rproj to the .gitignore

Latest commit 4138947 10 minutes ago

 **.gitignore**

Added .Rproj to the .gitignore

10 minutes ago

Help people interested in this repository understand your project by adding a README.

Add a README

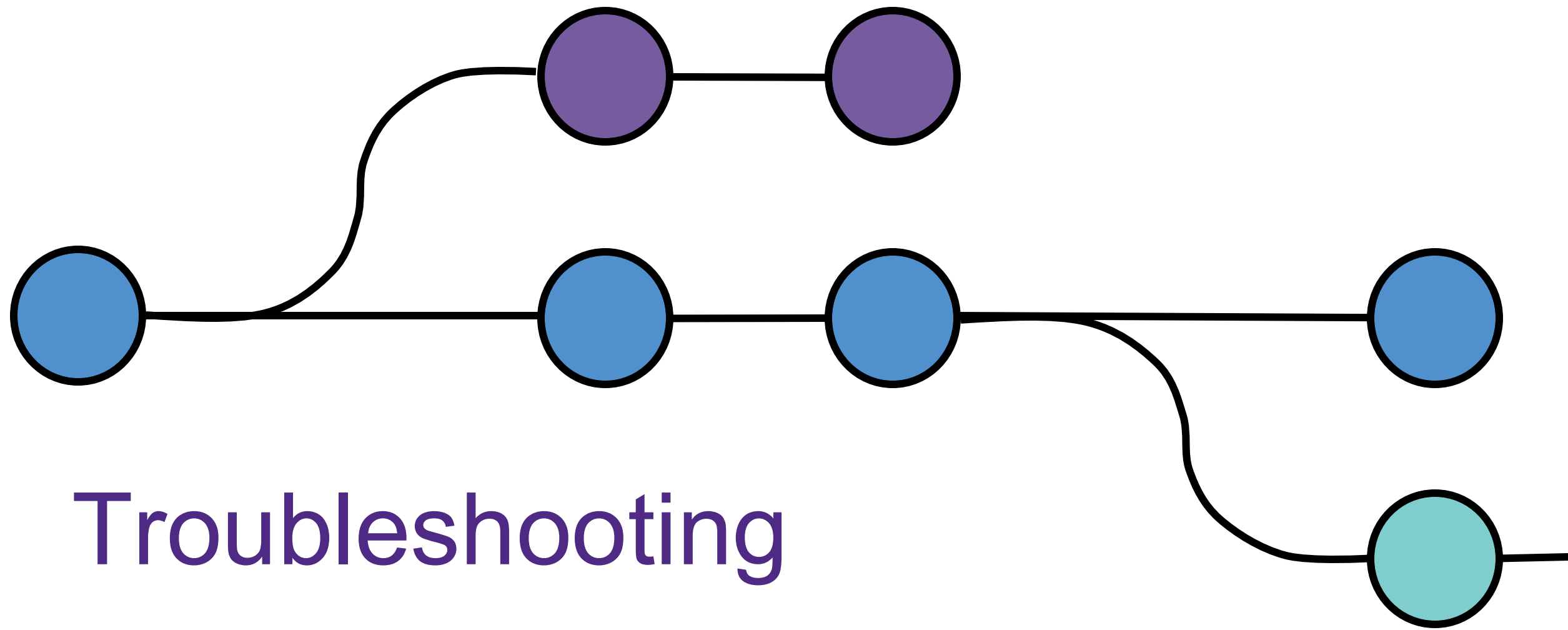
Practice:

- Make a new R script
- Save it to the R project directory
- Add some R code
- Do the steps necessary to get this file into your GitHub repo

Need help?

Workshop materials: <https://github.com/nuitrcs/git-RStudio>

- Request a consult: bit.ly/rcdsconsult
- Other Resources
 - [RCDS: Written Setup Instructions](#)
 - [RCDS: Other ways to set up Git repos in R](#)
 - What git is and how it works: [Git for Humans](#)
 - Why you should use git: [Academic benefits of using git](#)
 - How to use git with R and RStudio: [Happy git with R](#)



Troubleshooting

Q: I don't see the git tab

A: Check that RStudio sees git

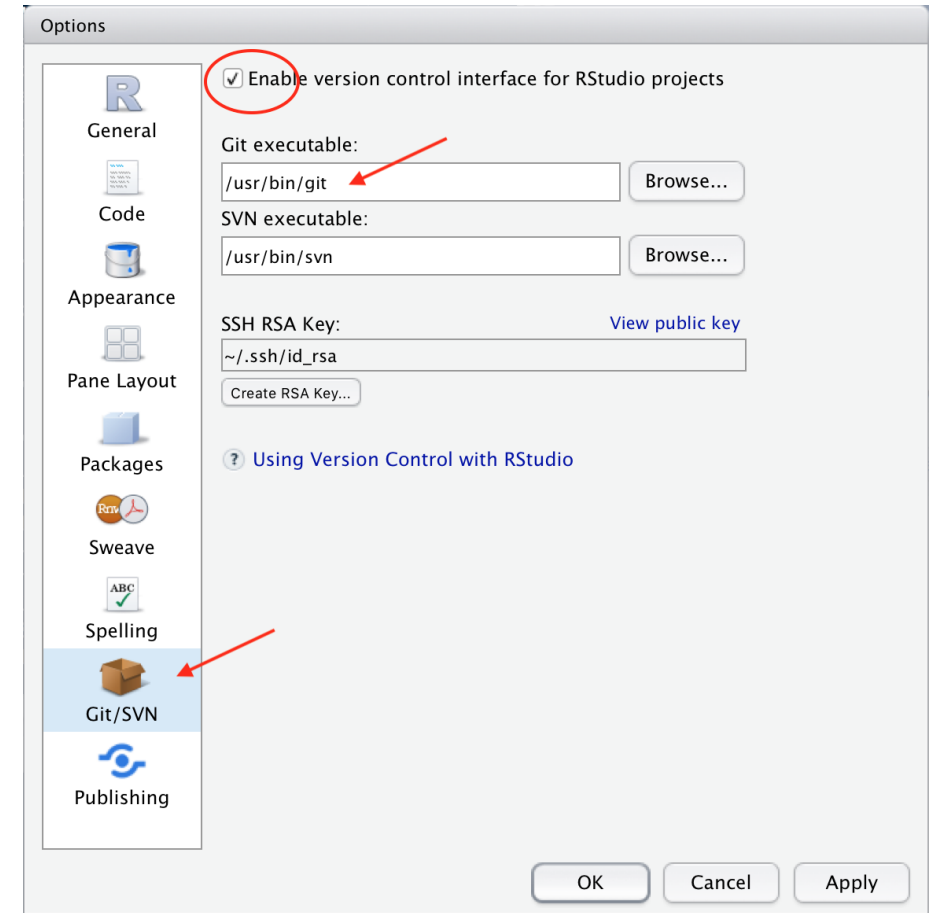
Mac: RStudio > Preferences

Windows: Tools > Global Options

Mac: type

which git

in the Terminal to get the path



Git workflow

